

RESEARCH ARTICLE

Decoding Emotional Dynamics: A Comparative Analysis of Contextual and Non-Contextual Models in Sentiment Analysis of Turkish Couple Dialogues

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ABSTRACT This paper introduces the “Couple Dialogue” dataset, specifically curated for conversational sentiment analysis in the Turkish language. It comprises 14,294 utterances from 118 dyadic conversations between couples, each annotated to capture sentiment transitions and interpersonal dynamics. Our study contrasts two distinct modeling frameworks—the Non-Contextual Model and the Contextual Model. The Non-Contextual Model analyzes utterances independently, typically overlooking the nuances of conversational dynamics and sentiment evolution. Within this framework, we conducted a detailed morphological analysis due to the agglutinative nature and rich morphological structure of the Turkish language, which included various word forms and negation morphemes crucial for sentiment representation. In contrast, the Contextual Model employs state-of-the-art Large Language Models (LLMs) such as BERT, GPT-3.5 Turbo, Llama 2, and GPT-4, alongside architectures like DialogueRNN. This model processes the sequential and relational aspects of dialogues through three approaches: prompt-based, fine-tuned, and embedding-based methods, particularly enhanced with fine-tuning and advanced embedding techniques (utilizing pre-trained and fine-tuned Turkish BERT). The Contextual Model substantially outperforms the Non-Contextual Model, showing a 9.98% improvement in Weighted F1 scores validated by statistical tests. Our work not only pioneers the use of LLMs in Turkish conversational sentiment analysis but also underscores the critical importance of contextual understanding in capturing complex emotional cues in couple interactions. This study sets a robust benchmark for future explorations into sentiment analysis within linguistically rich contexts.

INDEX TERMS Contextual model, conversational sentiment analysis, couple dialogue dataset, DialogueRNN, fine tuning, GPT, Llama 2, LLMs, morphological analysis, non-contextual model, Turkish language.

I. INTRODUCTION

Conversational sentiment analysis focuses on understanding the underlying sentiment throughout a conversation among

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participating individuals. This task requires the modeling of conversational contexts capable of identifying sarcasm, irony, and other complexities that cannot be discerned from individual utterances alone. Furthermore, analyzing sentiment based on conversations requires detecting changes in sentiment throughout the conversation. However, modeling

the flow of conversation and the dynamics of sentiment shifts poses a more intricate and challenging task. It is essential to overcome the challenge posed by multiple meanings of words depending on the context and to accurately detect ambiguity and sarcasm within the conversation's context.

Any form of conversation involves self and interpersonal dependencies [1], where self-dependency pertains to the relationships existing within an individual's own utterances, and interpersonal dependency refers to the relationships between multiple participants. Modeling self-dependency necessitates an understanding of the individual's prior statements and the tracking of changes in the individual's thoughts throughout the conversation. Modeling interpersonal dependencies requires awareness of the distinct speakers, the sentiments of each participant, and the impact of one participant's utterances on the sentiments of others. Self and interpersonal dependencies in contextual models can be modeled using long short-term memory networks [2], recurrent neural networks, transformers, and attention mechanisms [3], all of which are capable of retaining contextual information from previous utterances.

The Table 1 illustrates an interaction example between spouses. As seen in the dialogue, the female displays accusatory behavior due to her partner's careless attitude, leading to increased aggression during the conversation. However, while the male spouse initially maintains a calm demeanor, he becomes emotionally influenced by his wife, ultimately responding with anger towards the end. In this scenario, the female spouse maintains her harsh attitude without deviation, demonstrating self-dependency, while the male spouse is affected by her, exhibiting interpersonal dependency.

Conversational sentiment analysis is essential, as it enables the comprehension of the emotional flow, contextual shifts, and participant interactions within a conversation. Understanding the sentiment of the conversation is crucial for various applications including customer service analysis, social media monitoring, market research, and opinion mining for decision-making and determining necessary actions. While many studies focus on understanding the sentiment dynamics using traditional sentiment analysis approaches, such as analyzing individual words (Word Level) [4], sentences (Sentence Level) [5], or aspects [6] independently, modeling interaction dynamics remains an underdeveloped area of research.

Recently, a couple of conversational datasets, namely, IEMOCAP [7], DailyDialog [8], MELD [9], Persuasion for Good [10], and ScenarioSA [11], have been created to support emotional or sentimental conversational models in English. As far as we know, no annotated dataset is specifically designed for conversational sentiment analysis available in the Turkish language.

To address this gap, we initially compiled a conversational dataset in Turkish, annotated with sentiment labels, for the purpose of conducting research on conversational sentiment analysis. This dataset was named the Couple Dialogue

dataset. Our dataset comprises 118 dyadic couple conversations, encompassing a total of 14,294 individual utterances. These conversations were collected from heterosexual couples in Turkey, with the two speakers identified as "F" and "M" within each dialogue. Each utterance was manually annotated with its corresponding sentiment polarity, with "0" denoting a negative sentiment, "1" denoting a neutral sentiment, and "2" denoting a positive sentiment. As a result of privacy constraints, the data cannot be made publicly accessible.

The dataset stands out in several key aspects. Firstly, it represents a pioneering effort in the field of research on couple interaction within the Turkish context. Secondly, it stands as the most extensive collection of couple-generated utterances in the Turkish language that has come to our attention, with annotations included in a more meticulous way. Thirdly, the Couple Dialogue data set is the first data set collected and annotated aimed at facilitating research in the domain of conversational sentiment analysis. Fourthly, the Couple Dialogue dataset successfully captures the dynamics of interaction between couples, depicting the emotional changes of each speaker throughout the dialogue. Additionally, the dataset has been carefully annotated using a rule-based approach, minimizing subjectivity to the greatest extent possible. Our team dedicated considerable time and effort to ensure the precision of the labeling process. Finally, we conducted an inter-annotator agreement study and presented relevant statistical analyses to validate the reliability of the sentiment annotation procedure.

LLMs including BERT [12] and GPT-2 [13] have revolutionized natural language processing by improving the understanding of context and semantics in text. This transformation continued with the advent of models like GPT-3 [14], GPT-4 [15], and the open-source Llama 2 [16], which exhibit the ability to generate human-like text and manage a variety of tasks through zero- or few-shot learning [17]. The recently released Llama 2 by Meta GenAI, trained on two trillion tokens from publicly available sources, encompasses parameter counts from 7 billion to 70 billion. GPT-3, developed by OpenAI, comprises 175 billion parameters, while GPT-4, a multimodal model, could have more than 175 billion parameters, likely trained on trillions of tokens. This extensive training allows LLMs like Llama 2, GPT-3, and GPT-4 to excel at capturing context, understanding subtle language nuances, and replicating human dialogue patterns. Consequently, these models are proficient in detecting slight variations in tone and sentiment, making them highly effective for conversational sentiment analysis [18], [19], [20], [21], [22], which in turn provides valuable insights for businesses in customer service and market analysis.

LLMs can also employ natural language prompts [23], [24] to broaden their applications beyond specific fine-tuning for supervised tasks within transfer learning. Rather than collecting labeled training data for fine-tuning, users can instruct the model to achieve specific outcomes, such as asking it to "Classify the sentiment in these reviews as

TABLE 1. Example script from the Couple Dialogue dataset illustrating interactions between spouses, where “M” denotes male and “F” denotes female. Sentimental changes and shifts are indicated: “2” denotes positive, “0” denotes negative, and “1” denotes neutral.

Speaker	Utterance Text
F:	Mesela spora çok gidiyorsun, gitme spora! [0] (For example, you go to the gym a lot, don't go to the gym!)
M:	Abartıyorsun! Haftada iki gün gidiyorum. [0] (You're exaggerating! I go two days a week.)
F:	Arkadaşların çağırdığı zaman gidiyorsun, ben dediğim zaman gelmiyorsun! [0] (You go when your friends call, you don't come when I say so!)
M:	Örnek ver. [1] (Give an example.)
F:	Düşünüyorum. [1] (I am thinking.)
M:	Mesela ne aradılar da gittim? [1] (For example, why did they ask me to go?)
F:	Düşünüyorum. [1] (I am thinking.)
M:	Mesela ne zaman aradılar da gittim? [1] (For example, why did they ask me to go?)
F:	Düşünüyorum. [1] (I am thinking.)
M:	Bulamadın değil mi? (laughing) [2] (You couldn't find it, could you?)
F:	Geçenlerde X aradı, çıktın gittin! [0] (Recently, X called, you left!)
M:	Arkadaşım ile görüşmeye hakkım yok mu benim? [0] (Don't I have the right to meet my friend?)
F:	[0] (Recently, X called, you left!)
F:	Benden daha çok vakit geçirmeye hakkın yok! [0] (You have no right to spend more time than me!)
M:	Belki de bu tavırlarından dolayı uzaklaşmak istiyorum! [0] (Maybe I want to move away because of this attitude!)
F:	Hemen bahanenî buldun... [0] (You immediately found your excuse...)

either positive, negative or neutral.” This paper investigates the potential of LLMs for sentiment analysis through a prompt-based approach using GPT-4 and the chat version of the Trendyol LLM model, hosted on Hugging Face. The Trendyol LLM model,¹ developed by the Trendyol group, is a generative model based on the Llama 2 7B architecture, fine-tuned on 180,000 instruction sets, primarily in English and Turkish. The study utilized both instruction-based and few-shot prompting with GPT-4 and Llama 2. For GPT-4, only instruction-based prompting was employed, while Llama 2 utilized both instruction-based and few-shot prompting approaches. The effectiveness of these approaches was measured using F1 scores to assess the models’ performance in sentiment analysis.

Fine-tuning (FT) is a predominant method for customizing pre-trained language models for specialized tasks like dialogue systems [25]. This approach involves additional training on smaller, task-specific datasets to enhance model performance across various applications. However, fine-tuning can sometimes reduce a model’s generalization capabilities [26]. Anil et al. [27] suggest that incorporating multiple in-context examples during fine-tuning can help maintain text generalization. This study fine-tuned the Couple Dialogue dataset using models like BERT, GPT-3.5

Turbo, and Llama 2, with Llama 2 also undergoing few-shot fine-tuning to provide a comparative analysis of different fine-tuning methodologies.

LLMs generate embeddings for each token, which encapsulate both semantic and syntactic properties. The embedding layer converts numerical identifiers into dense, fixed-size vectors, providing rich linguistic information. This framework allows BERT and GPT models to capture intricate language patterns, contributing to their success in various language tasks. In this study, tokenized words were processed through BERT and GPT models, with embeddings extracted from specified layers. The “text-embedding-3-small” model,² was used to generate embeddings for GPT, while the pre-trained Turkish BERT and a fine-tuned version of it, utilizing the Couple Dialogue dataset, provided additional embeddings. GloVe embeddings (Global Vectors for Word Representation) [28] were also employed, serving as inputs to train a DialogueRNN model for conversational sentiment classification tasks. The DialogueRNN framework [29] encodes speaker-level information by taking into account factors such as speaker identity, context, and sentiment from prior utterances. This approach effectively captures both self-dependency and interpersonal dependency, leading to improved context representation.

¹<https://huggingface.co/Trendyol/Trendyol-LLM-7b-chat-v0.1>

²<https://openai.com/blog/new-embedding-models-and-api-updates>

This study aims to demonstrate the efficacy of context-based approaches in conversational sentiment analysis, focusing specifically on the “Contextual Model.” Experiments were conducted using a variety of LLMs on the Couple Dialogue dataset. Our models are divided into three categories: prompt-based (using GPT-4 and Llama 2), fine-tuned (employing BERT, Llama 2, and GPT-3.5 Turbo), and embedding-based (including DialogueRNN with embeddings from pre-trained Turkish BERT, fine-tuned Turkish BERT, text-embedding-3-small, and GloVe). This Contextual Model utilizes methods that range from taking instructions via prompts to leveraging fine-tuned datasets and embedding-based representations for deeper analysis. To establish a baseline for comparison, we developed a “Non-Contextual Model” that processes utterances independently using a variety of embeddings like BoW, TF-IDF, Word2Vec, and pre-trained Turkish BERT through traditional machine learning and deep learning techniques. This approach highlights the contrast between non-contextual processing and the nuanced capabilities of contextual models. We hypothesize that the Contextual Model will significantly outperform the Non-contextual Model, with this hypothesis to be rigorously tested through comparisons of F1 scores, providing a robust evaluation of both methodologies.

Next, in the experiments dedicated to the Non-Contextual Model, we evaluated the performance of classifiers by comparing the contextual embedding of BERT with non-contextual word embeddings, including TF-IDF (Term Frequency-Inverse Document Frequency) and Bag of Words (BoW), as well as word embeddings with less contextual information, such as Word2Vec and GloVe. This comparison was designed to assess the Non-Contextual model’s effectiveness relative to the contextual embedding, using the same Couple Dialogue data, and it revealed significant performance differences.

Furthermore, given the agglutinative nature and rich morphological structure of the Turkish language, we conducted a detailed morphological analysis specifically as part of the Non-Contextual Model experiments. This analysis included various word forms and negation morphemes, along with other morphemes that play a critical role in sentiment representation. We compared classifier performances based on Weighted F1 scores. Additionally, experiments involving the inclusion and exclusion of stopwords and punctuation marks provided insights into their impact on sentiment labeling.

Finally, we developed a deep neural network architecture that strikes a balance between complexity and performance. This architecture is less complex than the DialogueRNN model yet more sophisticated than conventional classifiers. It incorporates BERT and GloVe embedding layers, providing a middle ground between the Non-Contextual and Contextual Models.

In this study, we have the following research objectives:

i) To improve the success rates of sentiment analysis by transitioning from the Non-Contextual Model, which serves

as the baseline, to the Contextual Model, utilizing the Couple Dialogue data.

ii) To enhance the success rates of sentiment analysis by applying additional preprocessing and morphological analysis to the Non-Contextual Model using the same dataset.

To achieve these goals, the following research questions are addressed:

- **RQ1:** Does the Contextual Model consistently outperform the Non-Contextual Model on the Couple Dialogue data?
- **RQ2:** What is the most efficient prompt structure that would be understandable for both Llama 2 and GPT-3.5 Turbo in the context of conversational sentiment analysis on the Couple Dialogue data? (This question arises because, to the best of our knowledge, a suitable prompt structure for this task has not been documented in the literature.)
- **RQ3:** Can our conversational sentiment analysis task, which is based on the Turkish language, be effectively achieved by Llama 2 and GPT-3.5 Turbo, both of which are primarily pre-trained in English?
- **RQ4:** How much improvement do fine-tuned models offer over prompt-based approaches for this conversational sentiment analysis task?
- **RQ5:** Do contextual embeddings outperform less sophisticated vector representations for the Couple Dialogue data?
- **RQ6:** What are the effects of preprocessing and morphological analysis on a language with rich morphology, like Turkish, in enhancing sentiment labeling success rates?
- **RQ7:** Does the use of a more complex neural network architecture in the Non-Contextual Model help close the performance gap with the Contextual Model?

The key contributions of the research outlined in this paper can be summarized as follows:

- 1) We curated a dataset of couple interactions called the “Couple Dialogue” dataset, designed specifically for conducting conversational sentiment analysis in the Turkish language. Along with this dataset, we provided statistical analysis and demonstrated the reliability of the data collection and agreement procedures.
- 2) In this study, we developed and compared two types of frameworks: Non-Contextual, which serves as our baseline, and Contextual. We introduced a novel approach by employing contextual modeling techniques through instruction-based models, fine-tuned models, and embedding-based models, including the DialogueRNN architecture, specifically tailored for a Turkish dataset designed for conversational sentiment analysis. Our preliminary results indicate that the Contextual Model significantly outperforms the Non-Contextual Model, achieving statistically significant improvements of 9.98% in Weighted F1 scores with conventional machine learning classifiers and 6.51% with deep neural network architectures, as confirmed by paired t-tests.

- 3) We demonstrated that the fine-tuned approach significantly outperformed the prompt-based approach, achieving a notable improvement of up to 31.12% in Weighted F1 score for our conversational sentiment analysis task. This enhancement was confirmed to be statistically significant through a paired t-test.
- 4) Our analysis revealed that incorporating specific morphemes significantly enhances sentiment classification success in Turkish, evidenced by improvements of up to 9.89% in Weighted F1 scores. A majority of our tested pairs showed statistically significant enhancements, validating the effectiveness of these morphological features. We conducted selective paired t-tests across different classifiers and preprocessing settings, which supported the positive impact in most cases, though not universally. This underscores the overall efficacy of integrating morphemes in sentiment analysis while highlighting variability in results depending on the preprocessing context.
- 5) In our study, we illustrated that BERT-based utterance features consistently outperform other used vector embedding models within both Non-Contextual and Contextual frameworks. In the Contextual Model, the improvement was particularly notable, with a significant increase of 14.46% in Weighted F1 scores, as confirmed through statistical testing. In the Non-Contextual Model, BERT achieved an improvement of up to 7.42% and demonstrated statistical significance against all competing embeddings except for GloVe, where the difference did not reach statistical significance.

The rest of this paper is structured as follows: Section II provides a comprehensive review of related work in the field. Section III introduces the Turkish conversation dataset along with sentiment labels. In Section IV, we describe our proposed approach in detail. Section V explains the types of utterance features used in our experiments, while Section VI outlines the classifier models employed. Section VII is dedicated to the empirical evaluation of results, presenting a detailed analysis and discussion of our findings. Section VIII responds to our research questions addressed in the Introduction section. Section IX outlines potential future research directions and provides concluding remarks.

II. RELATED WORK

A. EARLY STUDIES IN SENTIMENT ANALYSIS

The field of sentiment analysis has remained prominent within natural language processing tasks. Early investigations in sentiment analysis concentrated on various levels, including sentence-level [30], document-level [31], aspect-level [32], and phrase-level approaches [33]. Besides that, some researchers implemented sentiment analysis-related tasks such as multimodal sentiment analysis [34], domain adaptation [35], sarcasm detection [36], [37], and the identification of biases in sentiment analysis [38], [39]. Recently, studies applying neural architectures, such as those

in [40], [41], [42], and [43] have overshadowed conventional machine learning algorithms.

B. STUDIES IN CONVERSATIONAL SENTIMENT ANALYSIS

Sentiment analysis in conversations has become a popular trend. Poria et al. [44] used contextual LSTM architecture that receives the sequence of utterances in a video to focus on determining sentiment. Hazarika et al. [45] proposed gated recurrent units to model the histories using audio, visual, and textual features to identify utterance-level emotions in videos of dyadic conversations. However, the mentioned studies are based on dyadic conversations. To model self and interspeaker emotional influences in memory hops, Hazarika et al. [46] introduced an Interactive Conversational Memory Network (ICON) and incorporated language, visual, and audio features in utterance videos. Their approach became an initial framework for multi-party conversations.

Majumder et al. [29] proposed the DialogueRNN framework which is scalable for more than two speakers. Their approach models three gated recurrent units (GRU); specifically, the global GRU modifies the contextual state, the party GRU updates the speaker or listener state, and finally, the emotion GRU handles emotion classification. In this study, the authors used convolutional neural networks (CNN) to extract textual features.

Zhong et al. [47] incorporated both contextual information and external knowledge in textual conversations and used two types of architectures, namely, hierarchical self-attention and cross-attention architectures to detect emotions. Ghosal et al. [48] constructed a dialogue graph convolutional network (DialogueGCN) with the purpose of capturing interactions among participants. Lian et al. [49] designed a conversational transformer network (CTNet) that combines a bi-directional GRU layer with a multi-head attention mechanism to capture important utterances extracting multimodal features.

Jiao et al. [50] constructed a network including Hierarchical Memory Network (HMN) and Attention GRU (AGHMN), where the HMN part improves the features of utterances and makes interaction among past utterances, and the Attention GRU part provides better memory summarization and comprehensive context. Zhang et al. [51] developed an interactive LSTM model for understandability, credibility, and influence interactions between speakers for conversational sentiment analysis. Besides that, they created a high-quality English conversational dataset, namely ScenarioSA which has been made publicly available to compensate for the lack of a conversational data set.

Zhang et al. [52] proposed a novel quantumlike multimodal network (QMN) approach that incorporates a quantum theory and a long short-term memory network for representing the dynamics of intra and inter-utterance interactions, utilizing both text and image features. Ghosal et al. [53] modified the existing LSTM and DialogueRNN baselines for five different dialogue understanding tasks using a pre-trained GloVe embedding and a pre-trained Roberta model [54].

We conducted experiments with our Contextual Model using the DialogueRNN model on Couple Dialogue data with pre-trained Turkish BERT, fine-tuned Turkish BERT, text-embedding-3-small, and GloVe embeddings, and we presented the results.

In Shou et al. [55], DSAGCN, a conversational affective analysis model is proposed, relying on dependent syntactic analysis and employing a graph convolutional neural network architecture. Their model demonstrated better average accuracy and F1 values for happiness and anger emotions for IEMOCAP [7] and MELD [9] datasets. In Wei et al. [56], the proposed model integrates the context information with the current utterance and then employs a two-channel feature extractor (TFE) to obtain emotional features. When the authors conducted an efficiency analysis and compared their model with two recent strong baselines namely DialogueRNN [29] and DialogueGCN [48], they illustrated that the suggested method surpassed those models in convergence capacity, the count of trainable parameters, and training duration. In the work of Tu et al. [57], a context- and sentiment-aware framework called Sentic GAT was introduced. This framework utilizes a hierarchical multi-head attention (HMA) mechanism to depict relationships among concepts in contextual dialogue, followed by the modeling of context- and sentiment-aware Graph attention (CSAGAT) in capturing dependencies between the current utterance and associated external concepts.

C. LLMS IN GENERAL AND CONTEXTUAL SENTIMENT ANALYSIS

Krugman et al. [18] presented an in-depth analysis of the effectiveness of Large Language Models (LLMs) such as GPT-3.5, GPT-4, and Llama 2 in sentiment analysis, a key element of marketing research for decoding consumer sentiments. Despite the zero-shot learning capabilities of LLMs, their performance can match or even surpass that of traditional transfer learning models in sentiment classification accuracy, influenced by factors such as text complexity and data origin. Additionally, the study highlights the superior explainability of LLMs, particularly Llama 2, in offering insights for method selection in an era dominated by generative AI. Notably, their research focused on binary and three-class sentiment classification, which differs from conversational sentiment analysis tasks.

Buscemi et al. [19] assessed the performance of leading LLMs, including ChatGPT, Gemini, and LLaMA2, in handling ambiguous and nuanced texts across ten languages. Their findings reveal that while LLMs like ChatGPT and Gemini generally interpret ambiguous scenarios effectively, they also exhibit significant biases and inconsistent performance, which varies across different languages and models. The outcomes, validated against human benchmarks, emphasize the need for continuous enhancements in LLM algorithms and training data to improve their accuracy, interpretability, and applicability in diverse linguistic and

cultural contexts. Additionally, their analysis was based on responses obtained through carefully focused prompts.

Zhang et al. [20] addressed the limitations of LLMs in financial sentiment analysis, particularly their struggles with numerical sensitivity and contextual understanding, by introducing an instruction-tuned model called Instruct-FinGPT. This model utilizes the extensive knowledge of pre-trained LLMs, specifically LLaMA, and applies instruction tuning to enhance its capabilities in financial contexts. It significantly outperformed both general-purpose LLMs and advanced supervised models with minimal instruction data. This novel approach underlines the critical role of numerical sensitivity and contextual awareness in financial NLP, offering a promising direction for future advancements in the accurate interpretation of financial sentiments.

Carvalho et al. [21] explored sentiment analysis applied to dialogues within Portuguese customer-support interactions, assessing the impact of incorporating various levels of contextual information, including the number and speaker of preceding utterances. By experimenting with a range of classification techniques from traditional methods to advanced models like fine-tuned BERT and Few-Shot Learning with GPT-3 and OPT, their findings highlight the variable significance of context in dialogue-based sentiment analysis. Context proves particularly critical in scenarios involving human-computer interaction and brief human-human exchanges focused on the client, thereby enhancing both the performance and the relevance of the dialogue systems in practical settings. This study represents one of the relatively rare examples of using LLMs for conversational sentiment analysis, emphasizing the importance of contextual information. In contrast to this study, our approach involves obtaining the entire context from the current conversation to provide a richer context, enabling the corresponding model to learn from previous interactions. We applied this comprehensive context logic across embedding-based, prompt-based, and fine-tuning-based approaches.

Kheiri et al. [22] conducted a comprehensive examination of Generative Pretrained Transformer (GPT) methodologies in sentiment analysis using the SemEval 2017 - Task 4 dataset. They evaluated three distinct approaches: prompt engineering with GPT-3.5 Turbo, fine-tuning GPT models, and an innovative embedding classification strategy. Their findings reveal that GPT-based methods outperformed existing state-of-the-art models by more than 22% in terms of F1 score, demonstrating their superior capability in handling complex aspects of sentiment analysis, such as contextual understanding and sarcasm detection. This research not only underscores the advanced performance of GPT models in sentiment analysis but also contributes novel insights into their application, facilitating a deeper understanding of linguistic nuances in automated sentiment assessment. For our task, we employed similar models to those used by Kheiri et al. However, their study focused solely on GPT-based models that performed three-class classification

on a Twitter dataset, which is distinct from our task of conversational sentiment analysis.

D. STUDIES IN OTHER MORPHOLOGICALLY RICH LANGUAGES

There is some work employed in other morphologically rich languages for the sentiment classification task. In the study by Abdul-Mageed et al. [58], inflectional and derivational suffixes were utilized in sentiment classification tasks for Arabic, significantly improving F1 scores when combined with SVM algorithms compared to using only stems. Joshi et al. [59] developed a three-stage system for extracting sentiments from Hindi reviews, employing supervised algorithms with annotated corpora, machine translation to English when corpora were unavailable, and sentiment lexicons for classification via majority voting. Yang and Chao [60] classified Chinese movie reviews by leveraging seed words and mono-syllabic morphemes, effectively using specific suffixes as features without sentiment resources. Medagoda [61] utilized machine learning approaches to classify Sinhala reviews, analyzing the impact of selecting appropriate adjectives and adverbs on classification success and generating polarity scores through different weighting schemes. Furthermore, Medagoda [62] proposed various methods for generating sentiment lexicons for Sinhala, including cross-lingual methods, incorporating morphological features, and a graph-based approach, integrating rule-based methods with negations and intensifiers, claiming applicability to other morphologically rich languages.

E. OTHER RELEVANT WORK

Zhang et al. [63] proposed a multi-task learning framework with an attention mechanism for semantic and instance segmentation in coastal urban spatial perception. This study highlights how attention mechanisms can improve model performance in complex tasks by effectively managing various subtasks simultaneously. The framework's ability to handle multiple tasks could inspire advancements in conversational sentiment analysis by integrating attention mechanisms to focus on different aspects of dialogue. In our work on conversational sentiment analysis for Couple Dialogue data, we utilized the DialogueRNN model within our embedding-based approach, which belongs to our Contextual Model framework. DialogueRNN uses an attention mechanism to dynamically focus on relevant parts of the conversation history, allowing the model to better understand the context and dependencies between utterances. This is achieved by computing attention scores for each past utterance, normalizing them to obtain attention weights, and creating a context vector that influences sentiment or emotion prediction. By effectively managing the conversation's context, this mechanism enhances the model's ability to predict sentiments/emotions accurately.

In the realm of cross-subject emotion recognition, Xiaopeng Si and colleagues [64] developed a novel

brain-computer interface using functional near-infrared spectroscopy (fNIRS) and a dual-branch joint network (DBJNet). The study introduced deep learning technology to fNIRS for the first time, demonstrating a significant improvement in emotion decoding accuracy. This work is notable for its exploration of cross-subject emotion recognition, showcasing how neural activity data can be leveraged for more accurate emotion classification, a concept potentially applicable to understanding neural correlates in conversational sentiment analysis.

Zhu's paper [65] focuses on developing an intelligent assistant system equipped with emotion recognition capabilities to enhance emotional intelligence and provide personalized services. The system uses multimodal data, including speech, text, and facial expressions, to accurately identify users' emotional states and tailor services accordingly. By analyzing user behavior and emotions, the assistant can offer more relevant and satisfying interactions, improving overall user experience. This approach emphasizes the integration of emotional intelligence in real-time sentiment analysis applications, enhancing the system's ability to respond empathetically and effectively to user needs.

Finally, the SiMaLSTM-SNP model by Gu et al. [66] combines Siamese networks and membrane computing for semantic relatedness learning. This work demonstrates that SiMaLSTM-SNP, which integrates word embeddings, multi-head self-attention, and LSTM-SNP, outperforms several classical and modern approaches in capturing the semantic distinctions between sentences. This innovative approach provides insights into advanced neural architectures that could be adapted to enhance conversational sentiment analysis models by improving the understanding of semantic relationships within dialogues.

F. STUDIES IN TURKISH SENTIMENT ANALYSIS

1) THE IMPACT OF AGGLUTINATIVE NATURE OF TURKISH ON SENTIMENT ANALYSIS

Turkish, as an agglutinative language, presents unique opportunities and challenges for sentiment analysis due to its rich morphological structure. Understanding these linguistic features is crucial for evaluating the development and performance of sentiment analysis systems in Turkish.

a: ADVANTAGES: RICH MORPHOLOGICAL INFORMATION

The agglutinative nature of Turkish enriches morphological information and enables nuanced sentiment analysis. For example, the word "seviyorum" (I love) combines the root "sev" (love) with the suffix "-iyorum," indicating present continuous tense and first person singular. This morphological richness, as noted in studies like [67], allows sentiment analysis systems to capture subtle variations in sentiment, enhancing accuracy. Suffixes in Turkish also express context, relationships, and emotions such as respect or sarcasm, which are crucial for accurate sentiment detection [68].

b: CHALLENGES: COMPLEX MORPHOLOGICAL PARSING AND DATA SPARSITY

However, this complexity also introduces challenges. The extensive use of suffixes can result in highly complex word structures, making morphological parsing difficult and computationally intensive [69]. Additionally, as previously mentioned, the high number of possible word forms leads to data sparsity, complicating model training and increasing the risk of overfitting [70]. The large vocabulary size and the resulting high out-of-vocabulary (OOV) rate further complicate statistical language modeling [71], necessitating advanced methods to handle these challenges effectively.

These linguistic characteristics have been explored and leveraged in various studies, as seen in the following early works on Turkish sentiment analysis.

2) EARLY STUDIES IN TURKISH SENTIMENT ANALYSIS

Numerous recent studies have focused on Turkish sentiment analysis. In Aydın et al. [67], the authors conducted morphological analysis and utilized various word forms of the Turkish language to address the binary sentiment classification problem. Their framework encompassed supervised, unsupervised, and semi-supervised methodologies, along with a combination of these methods. Additionally, aside from utilizing the surface and root forms of words as features, they considered certain morphemes carrying more sentimental information than others, generating a partial surface form as a third form. They applied their proposed method to a dataset of movie reviews and a Turkish Twitter dataset, demonstrating significant improvements over the baseline. Similarly, we generated a partial surface form that includes the suffixes utilized in this study if they were present in the surface form of a specified word.

In Omurca et al. [5], the authors collected hotel reviews from a web source and created a semi-automatically annotated corpus at the sentence level for potential use in aspect-based sentiment analysis studies in the Turkish language. Dehkharghani et al. [68] implemented a sentiment analysis system encompassing sentence, aspect, and document levels, incorporating a range of comprehensive linguistic features. They achieved accuracies ranging from 60% to 79%. In Güler et al. [72], the authors generated eight different sets of embeddings based on Word2vec and FastText models, focusing on various morphological forms of words. They showcased the contribution of these embeddings by conducting comparisons in word analogy, sentiment analysis, text classification, and language modeling.

3) TRANSFORMER-BASED STUDIES IN TURKISH SENTIMENT ANALYSIS

Köksal et al. [73] introduced the BounTi dataset, comprising Turkish tweets centered around specific universities in Turkey. They manually annotated the data for the Sentiment Analysis task and demonstrated the Turkish transformer models' performances, including MBERT, XLM-Roberta, and

BERTurk, on the dataset. In Ozturk et al. [74], transformer-based models such as BERT, ConvBERT, and ELECTRA, along with conventional deep learning and machine learning algorithms, were employed on four Turkish datasets, and their performance rates were subsequently compared.

Acikalin et al. [75] conducted experiments on binary classification problems using Turkish movie and hotel review datasets. They utilized both a multilingual version of BERT and the main BERT model after generating translated texts from Turkish to English, subsequently comparing the results obtained. Guven [76] evaluated the accuracy performance of pre-trained models, namely, BERT, ALBERT, ELECTRA, and DistilBERT in the Turkish language, specifically on sentiment classification tasks using the hotel dataset and movie review dataset. Mutlu et al. [77] introduced a manually annotated Turkish Twitter dataset suitable for targeted sentiment analysis tasks. They conducted experiments using several variants of BERT-based models, demonstrating a significant enhancement in performance compared to the baseline.

In recent studies, [78] marks a pioneering effort in fine-tuning the BERT-based model, BERTurk, for four critical natural language understanding (NLU) tasks—Named-Entity Recognition, Sentiment Analysis, Question Answering, and Text Classification—specifically tailored for the Turkish language. Leveraging a standard BERT model trained with base settings, the research successfully outperformed existing baseline approaches in these tasks as evaluated using the Turkish Benchmark dataset. Furthermore, [79] study introduces ITUTurkBERT, a novel collection of BERT models fine-tuned for Turkish, a morphologically complex language. By examining the effects of various tokenization methods, particularly focusing on vocabulary size and morphological tagging, the research extends beyond typical tokenization strategies and assesses their impact on NLP tasks such as Named Entity Recognition, Sentiment Analysis, and Question Answering. Empirical results demonstrate that while increasing vocabulary size generally enhances performance, sentiment analysis does not follow this trend, suggesting the need for further study in this area.

4) STUDIES IN TURKISH SENTIMENT ANALYSIS BASED ON INTERACTION DATASET

Toh et al. [80] introduced a hybrid model designed to enhance the understanding of negative and positive emotions in the task of human-robot interaction. They curated a novel dataset comprising 13,500 question pairs, which were converted to vector forms using pre-trained Turkish BERT, Word2vec, and TF-IDF. The integration of polarity vectors, representing the negative and positive word scores in the question-answer text vector, resulted in a substantial increase in accuracy compared to previous models. In another study, [81] designed a chatbot aimed at promoting credit card services to customers, employing a Finite State Machine (FSM) structure for the Turkish Language. The sentiment

classification model's output determined transitions between predefined states. They compiled a dataset comprising 500 utterances, annotated the utterances, and presented sentiment classification scores using several algorithms.

In conclusion, it is evident that conversational sentiment analysis studies are advancing, encompassing various datasets such as modalities or multimodalities, effectively applying feature engineering techniques and deep learning architectures. However, further efforts are warranted in the realm of conversational sentiment analysis for the Turkish language. As far as we know, this study marks the initial endeavor to employ the sentiment analysis task on conversational Turkish text using the Transformer model. Additionally, the collected and annotated dataset is the first based on Turkish couple interactions. To the best of our knowledge, there has not been any gathered and tagged data including dialogue utterances for conversational sentiment analysis in Turkish.

The study conducted by Söğüt et al. [81] includes 500 Turkish utterances, employing sentiment labeling through utterance-based features such as the count of positive terms or the count of negative terms in a sentence. In another study, Kılıç et al. [82] utilized an annotated corpus of human-machine dialogues comprising 426 conversations and 5020 sentences to classify dialogue acts from the banking domain in the Turkish language. As depicted in Table 2, our work was conducted on a more extensive dataset, encompassing 118 dialogues comprising 14,294 annotated utterances.

Furthermore, this study represents the first attempt to model contextual dependency for the purpose of utterance-level dialogue comprehension in sentiment classification tasks using Turkish text. Despite the growing focus on aspect-level sentiment analysis [6], document-level sentiment analysis [32], and sentence-level sentiment analysis [5], the development of a dialogue-based sentiment detection system in Turkish has not been previously explored.

5) ADVANCEMENTS IN TURKISH NLP USING LLMs

Uludogan et al. [83] introduced TURNA, the first unified language model for Turkish capable of performing both natural language understanding (NLU) and generation (NLG) tasks. Developed using a diverse corpus that includes web data, scientific articles, and parliamentary transcripts—totaling 43 billion tokens—TURNA employs an encoder-decoder architecture and boasts 1.1 billion parameters. In comparative evaluations across thirteen datasets and eight tasks, TURNA demonstrated superior performance compared to existing multilingual models, matching or exceeding the Turkish-specific BERTurk in NLU tasks and outperforming mBART and mT5 in all classification tasks for datasets such as Product Reviews21, TTC490022, and Tweet Sentiments.

Turker et al. introduced VBART [84], the first Turkish-specific sequence-to-sequence Large Language Model (LLM), pre-trained on a comprehensive 135.7 GB corpus. Incorporating elements from BART and mBART

architectures, these models, available in Large and XLarge sizes, significantly surpass previous benchmarks in abstractive text summarization, text paraphrasing, title generation, question answering, and question generation, establishing new standards for Turkish NLP tasks.

TurkishBERTweet [85] presented the first large-scale pre-trained language model tailored for analyzing Turkish social media content, built using nearly 900 million tweets. Optimized for shorter input lengths to enhance inference speed, this model demonstrates superior performance in sentiment classification and hate speech detection compared to existing models like BERTurk and multi-lingual models. TurkishBERTweet also proves to be a cost-effective and computationally efficient solution, offering significant advantages for handling large-scale datasets. To support further research and application development, the model and associated LoRA adapters are made available under the MIT License, marking a significant contribution to the field of Turkish NLP and social media analysis.

Cam et al. [86] evaluated the efficacy of ChatGPT and various fine-tuned BERT-based transformer models in detecting Turkish hate speech, finding that ChatGPT performs comparably to these models and demonstrating its potential in this application. Denghan et al. [87] assessed ChatGPT's ability to detect hate speech in Turkish tweets using the SIU2023-NST dataset, achieving 65.81% accuracy in a few-shot setting compared to 82.22% for a supervised fine-tuned BERT model, highlighting BERT's greater suitability for natural language classification tasks like hate speech detection despite its smaller size.

Studies focusing on Turkish sentiment analysis are scarce. Our study introduces significant novelties, including being the first to examine the contextual framework for conversational sentiment analysis, considering speaker dependency, previous context, and sentiment dynamics. Uludogan et al., Turker et al., and TurkishBERTweet have developed specific LLMs for the Turkish language. This study is also the first to compare the efficacy of fine-tuning and prompt-based approaches using a Turkish dataset specifically created for conversational sentiment analysis, employing prominent and recent LLM models such as GPT-4, GPT-3.5 Turbo, and Llama 2.

III. CORPUS DESCRIPTION

A. PARTICIPANTS AND PROCEDURES

The data were gathered from a sample of 103 recently married heterosexual couples in Turkey, all of whom were experiencing their first marriages and did not yet have children. The length of time the couples had been married varied from one to 15 months, averaging 6.06 months ($SD = 3.43$). The age of participants varied from 20 to 48 years, with a mean age of 27.83 ($SD = 3.80$). Participants had an educational background of 16.36 years on average, with individual durations spanning from 8 to 26 years.

Recruitment strategies involved distributing flyers at various universities, municipalities, and organizations situated

in an urban area. Additionally, recruitment messages were disseminated on social networking sites and diverse email groups. People indicating their interest were then contacted via phone or email to provide them with comprehensive details about the study. After agreeing to participate, participants received an email with an ID number and a link to access the online survey. Prior to data collection, participants were briefed on the study's objectives and procedures, and informed consent was obtained.

B. AUDIO-VISUAL DATA

The audio-visual data were collected in the Relationship Research Laboratory at Özyegin University in Istanbul, Turkey. The data were collected in rooms equipped with video cameras. The data collection process followed the procedures outlined in the Couples Interaction Rating System (CIRS) created by Heavey et al. [88]. CIRS serves as an observational coding scheme used to assess the dynamics of marital interactions. It focuses on evaluating communication patterns, conflict resolution effectiveness, and emotional expressions during discussions, providing a structured approach to understanding relational dynamics.

Upon arrival at the lab, each spouse was instructed to choose a topic they viewed as significant in their relationship and on which they held differing opinions. After both spouses independently selected their topics and mutually agreed on each other's selections, the couple engaged in two 10-minute discussions, each focusing on one spouse's chosen topic. At the 10-minute mark, an assistant knocked on the door, signaling the couple to transition to the next (other partner's) topic. After completing the last 10-minute segment, the research assistant knocked once more to signal the end of the recording. The recordings where couples engaged in 10 minutes of problem discussion comprised the audio-visual data of the current study.

C. COUPLE DIALOGUE DATASET

Our goal for this work is to conduct sentiment analysis on the collected data outlined above. Initially, we transcribed each spoken corpus into written text in the Turkish language. To ease and accelerate the annotation process, we randomly selected 118 interactions out of the possible 206 interactions (two conversations for each of the 103 couples), resulting in 118 separate conversation texts. Subsequently, we annotated the sentiment and emotional labels for each utterance within every conversation, resulting in the generation of the Couple Dialogue dataset. In contrast to the approach in [51], we chose not to annotate the entirety of the conversation in this study; our focus lies solely on the utterance-level analysis.

Since we transcribed texts from the spoken corpus, many of these conversations could be characterized as informal, resulting in a limited vocabulary of 4656 unique words. The detailed statistics are available in Table 2.

As seen in Fig. 1, we observe a smaller sample size for positive samples, totaling 2083, rendering our Couple

TABLE 2. The key statistics from the Couple Dialogue dataset, detailing the composition and structure of conversations and utterances, including word counts and gender distribution.

Statistics	
The total number of conversations	118
The total number of utterances	14,294
The total number of words in all corpus	142,306
The total number of unique words in all corpus	4,656
The total number of male utterances	6,798
The total number of female utterances	7,496
Average words per conversation	1,205.9
Average words per utterances	9.95

Dialogue Dataset imbalanced. Given that the couples engaged in discussions to resolve disagreements within their marriage, this disparity in sample sizes can be considered reasonable. In contrast, the number of negative and neutral samples is comparable. Specifically, we have 6161 negative samples, whereas the total number of neutral samples amounts to 6050.

The distribution of sentiment labels regarding gender diversity is shown in Fig. 2 as percentages, indicating that female spouses developed a more negative attitude during conversations compared to male spouses. While the percentage of females demonstrating a negative attitude was 25.0%, this value was 18.1% for male spouses. The percentage of positive sample values remained at 6.6% for the female group, while male spouses had 8.0%. Both these statistics demonstrate that female spouses exhibited a greater tendency towards negativity in discussions aimed at resolving problems compared to their male counterparts.

D. DATA PREPROCESSING IN DATA COLLECTION

The speech data was directly converted into text, resulting in the transcribed format of the conversation containing numerous illegible characters and misspellings. A significant challenge during the collection process was the lack of sentence separation by punctuation marks in the converted text, making the transcripts complex and difficult to read. To enhance the clarity and coherence of the sentences, extensive effort was devoted to appropriately inserting punctuation marks and correcting misspellings. Punctuation marks were manually provided as we did not find any tool capable of meaningfully segmenting the joined text using punctuation marks in the Turkish language.

In addition to punctuation correction, we developed a custom autocorrect tool to address misspellings within the corpus, utilizing the Microsoft Office Spell Checker as part of our initial automated correction phase. While this integration automated some of the corrections, it was not entirely sufficient, necessitating further extensive manual review to ensure all remaining misspellings were accurately corrected. Therefore, we conducted an exhaustive manual review to ensure all remaining misspellings were accurately corrected. This two-tiered approach to text correction—initial automated corrections followed by manual rectification—was essential to achieve the high-quality, readable text necessary for our analysis.

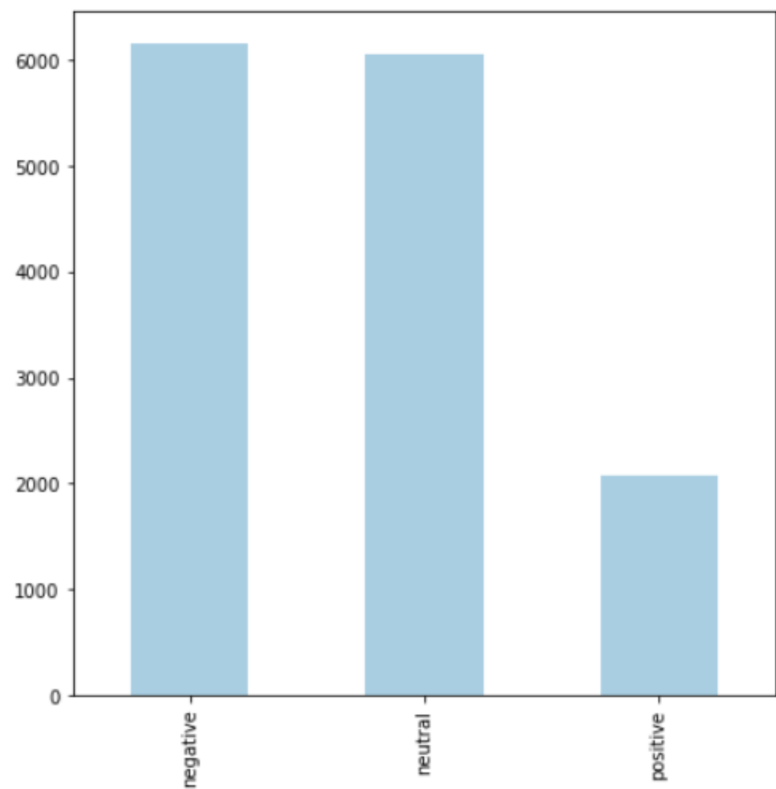


FIGURE 1. Distribution of positive, negative, and neutral sentiments in the Couple Dialogue dataset, illustrating the count of each sentiment type.

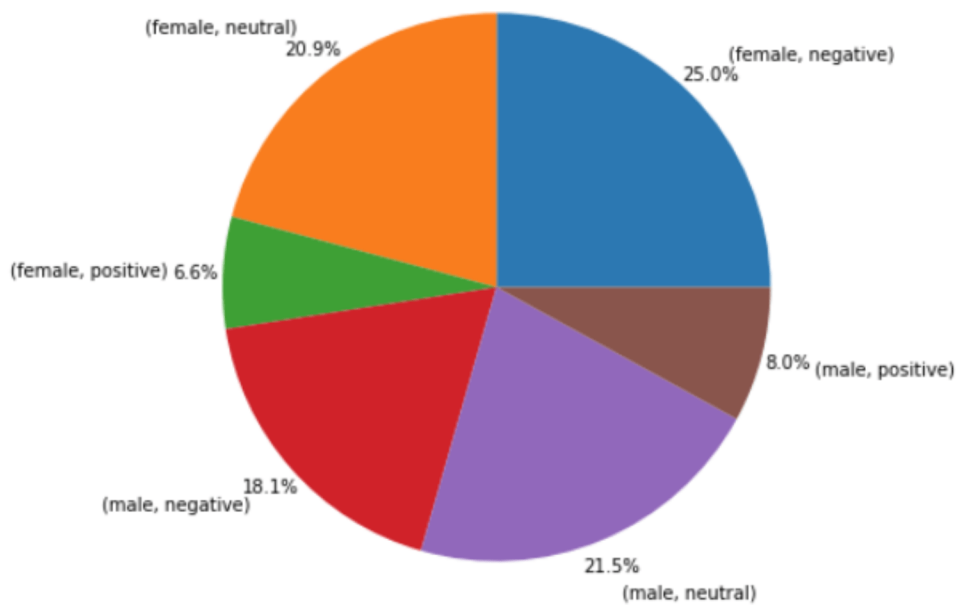


FIGURE 2. Distribution of positive, negative, and neutral sentiment labels across male and female speakers in the Couple Dialogue dataset.

E. ANNOTATION PROCEDURE

In the preprocessing phase, our dataset underwent manual annotation for both sentiment and emotional expressions.

Although the focus of this paper is on sentiment analysis, the emotional labels provided essential context for each interaction, aiding in the accurate classification of sentiments.

First, we annotated each utterance with one of seven emotional labels: 0 for “anger”, 1 for “no emotion”, 2 for “disgust”, 3 for “fear”, 4 for “happiness”, 5 for “sadness”, and 6 for “surprise.” Then, based on the emotion label, each utterance was assigned one of three sentiment labels: 0 for “negative”, 1 for “neutral”, and 2 for “positive.”

This dual-labeling strategy allowed for a streamlined sentiment classification process. For instance, utterances labeled as “anger” typically correlated with a negative sentiment. By identifying the emotional state first, annotators were able to assign sentiment labels more accurately and consistently. We have reserved a more detailed exploration of these emotional annotations for future research, as discussed in the Future Work section.

Initially, we annotated the dataset ourselves to establish a baseline for sentiment and emotion labels, which we consider as the ground truth labels. These ground truth labels were subsequently used in all experiments presented in this paper, ensuring consistency and reliability across the analysis.

To enhance the reliability of these ground truth labels, we engaged four volunteers, all experienced in sentiment analysis, to validate and refine our work. The labels provided by these annotators were specifically used to assess the reliability of our initial ground truth labels, rather than being directly used in the experiments.

To ensure consistency and minimize subjectivity among the annotators, we developed a comprehensive, rule-based guideline document grounded in the principles set forth by the Paul Ekman Group [89]. This document provided explicit criteria for recognizing and categorizing sentiments and emotions based on textual cues. While Paul Ekman’s work is renowned for identifying universal emotions through facial expressions and body language, our guidelines adapt these concepts to the conversational context by focusing on the language that typically reflects these emotions—*anger*, *disgust*, *fear*, *happiness*, *sadness*, and *surprise*—through specific lexical choices, tone, and contextual analysis. The detailed explanations for each emotion are given below:

- **Anger** is annotated when utterances include strong, confrontational words or phrases that suggest frustration or aggression. For example, “Why do you always mess this up?” could indicate anger, especially when combined with exclamation points or harsh adjectives.
- **Happiness** is identified through the use of positive expressions, exclamation marks, and affirming language that typically conveys joy or satisfaction. For example, “That’s fantastic! I’m thrilled to hear that.”
- **Sadness** involves words or phrases that express loss, longing, or disappointment. Phrases like “I really miss those days” or “I wish things were different” often convey sadness.
- **Surprise** is indicated by expressions that show astonishment or unexpected reactions, often marked by exclamatory phrases such as “Wow! I wasn’t expecting that!” or “How did that happen?”

- **Fear** can be deduced from expressions that reflect anxiety, worry, or concern. For instance, “What if it doesn’t work out?” or “I’m worried about this” suggests fear.
- **Disgust** is marked by words that express strong aversion or disapproval, such as “That’s horrible!” or “I can’t stand this.”
- **No Emotion** is identified when the text is factual or neutral, lacking any emotional words or phrases. An example might be “The meeting is scheduled for 3 PM.”

F. TRAINING AND CALIBRATION OF ANNOTATORS

During the annotation phase, annotators first assigned one of the seven emotional states to each utterance, followed by a sentiment label in separate sessions to prevent bias in immediate decision-making. The volunteers underwent an intensive training program that included reviewing case studies and participating in trial scoring sessions, where they practiced applying the guidelines to sample utterances. This training aimed to align their understanding and application of the labeling criteria. The guideline provided explicit criteria for categorizing sentiments linked to these emotional expressions. Firstly, we included general rules in the guideline as a pseudocode given below:

```
function detect_sentiment(emotion, context):
    if emotion == 'No Emotion':
        return 'Sentiment: Neutral'

    elif emotion == 'Happiness':
        return 'Sentiment: Positive'

    elif emotion == 'Surprise':
        # Context Check
        if context == 'positive outcome \
(e.g., a surprise party)':
            return 'Sentiment: Positive'
        elif context suggests concern for safety:
            return 'Sentiment: Negative'
        else:
            return 'Sentiment: Neutral'

    elif emotion in ['Anger', 'Disgust']:
        return 'Sentiment: Negative'

    elif emotion == 'Sadness':
        # Context Check
        if context suggests a resigned or \
matter-of-fact discussion:
            return 'Sentiment: Neutral'
        else:
            return 'Sentiment: Negative'

    elif emotion == 'Fear':
        # Context Check
        if context == 'immediate threat or loss':
            return 'Sentiment: Negative'
        elif context == 'mild or cautionary \
without immediate threat':
            return 'Sentiment: Neutral'

    else:
        return 'Sentiment: Neutral'
```


Our annotation guidelines not only delineate clear categorization rules but also address the complexity inherent in human emotions, especially when they manifest in layered or mixed forms. This is particularly challenging in conversational contexts where expressions such as sarcasm or understated emotions can alter the apparent sentiment of an utterance. The example below is integral to our guidelines, serving both as training material and as reference points to aid annotators in recognizing and distinguishing between nuanced emotional expressions. These cases often require careful consideration of both the lexical content and the broader conversational context to accurately determine the dominant emotional state. For instance, an utterance may initially seem to express happiness but upon closer examination, reveal underlying sarcasm or disappointment. Through detailed scenario-based training, our annotators learn to navigate such complexities, ensuring a more consistent and accurate application of the emotional and sentiment labels across our dataset. We included additional examples in Appendix A.

G. EXAMPLE: HAPPINESS VS. SARCASM

- **Dialogue Snippet:** “Harika, yine bir tartışma daha.” / “Great, another argument.”
- **Context:** Said after a series of arguments.
- **Annotation Challenge:** Determining whether this is an expression of genuine happiness or sarcasm.
- **Guideline Application:** The phrase implies dissatisfaction despite its positive wording, leading to the interpretation of sarcasm.
- **Final Sentiment:** If interpreted as sarcasm within a negative context, the sentiment is **Negative**.

We also showed examples of the concept of emotional blends, where multiple emotions might be present in a single utterance, reflecting the complexity of human interactions. For example, in a case study involving mixed emotions of happiness and surprise, annotators learned to distinguish primary from secondary emotions based on contextual cues such as preceding and following utterances. In the guideline, we incorporated a relevant example given below:

H. EXAMPLE: ANGER VS. NEUTRAL

- **Dialogue Snippet:** “Yine geç kaldın, bu gerçekten sinir bozucu!” / “You’re late again, this is really annoying!”
“Ama artık daha dikkatli olacağını biliyorum.” / “But I know you’ll be more careful from now on.”
- **Context:** Said during a discussion about repeated lateness.
- **Annotation Challenge:** Distinguishing between anger due to frustration and a neutral statement showing understanding.
- **Guideline Application:** Consider the tone and content of both sentences. The first sentence indicates anger with strong, confrontational words (labeled as Anger), while the second sentence shows a neutral tone with a hopeful and understanding statement (labeled as Neutral).

- **Final Sentiment:** If the anger is more prominent and intense, the final label should be **Anger** (Negative). Otherwise, if the neutral tone significantly balances the overall sentiment, it can be **Neutral**.

Other relevant examples were included in Appendix B.

Guideline documents were meticulously prepared in Microsoft Word format and introduced to annotators during the Initial Workshop, where each section of the document was thoroughly explained. These documents included definitions, examples, and instructions for handling edge cases. Following the introductory session, hands-on practice was conducted using real and simulated examples drawn from the Couple Dialogue dataset to expose annotators to practical scenarios they would encounter in the data. These practice sessions helped bridge the gap between theoretical knowledge and real-world application by allowing annotators to engage directly with complex or ambiguous cases. Annotated utterances were shared in CSV format via Google Drive, where annotators could download, annotate, and re-upload their files. Each annotator worked independently initially to ensure a diverse range of interpretations.

To minimize erroneous labeling and promote consistency, calibration meetings were regularly held. During these sessions, challenging cases where annotators disagreed were discussed collectively to reach a consensus. This collaborative approach ensured a high level of accuracy and alignment with the project’s standards.

I. AGREEMENT ASSESSMENT

Given our limited resources, we engaged four volunteer annotators to validate and refine our initial baseline annotations, which we had previously established ourselves. To ensure a high level of reliability, we focused on annotating 10% of the dialogues in detail. Specifically, the volunteers annotated the dialogues of 12 couples, totaling 1609 utterances. The reliability of these volunteer annotations was assessed using several statistical measures, with our initial annotations serving as the ground truth for comparison. Among the 1609 utterances, there were 656 negative samples, 631 neutral samples, and 322 positive samples.

We calculated the average agreement score by comparing the labels assigned by the five annotators (four volunteers plus our initial label). For each sample, if three or four of the volunteer annotators agreed on a specific label (positive, negative, or neutral), that label was considered the “agreed” label. This agreed label was then compared with the ground truth label, and if they matched, it was counted as an agreement. The overall agreement score, reflecting the percentage of matching agreed labels with the ground truth, was approximately 74.02% for all annotated utterances. Additionally, we assessed the agreement by categorizing sentiments into subjective (positive and negative) and objective (neutral) labels. For this categorization, we obtained an average agreement score of 78.06%.

The difference of 4.04% between the general agreement score and the subjectivity/objectivity agreement score

highlights the challenges in distinguishing between subjective and objective sentiments. This difference may be influenced by the inherent difficulty in consistently identifying subjective sentiments, especially given the relatively low number of positive samples. The higher agreement on the broader categorization of subjective versus objective suggests that annotators found it somewhat easier to agree on the general tone (subjective vs. objective) than on the specific sentiment (positive, negative, or neutral).

Further, we employed Fleiss' Kappa, a statistical measure that evaluates the consistency of agreement among multiple raters, accounting for the possibility of agreement occurring by chance. This metric is ideal for analyses involving more than two annotators, making it more suitable for our study with five annotators than Cohen's Kappa, which is designed for just two raters. The formula for Fleiss' Kappa is defined as:

$$\kappa = \frac{P_0 - P_e}{1 - P_e} \quad (1)$$

where P_0 is the observed agreement among raters, and P_e is the expected agreement by chance.

To calculate P_e , we first determine the proportion of each sentiment label assigned by all raters over all utterances, then square these proportions and sum them to reflect the likelihood of random agreement if raters were guessing based on the distribution of labels.

Applying this formula, we achieved a κ score of 59.4%, which places our agreement level in the "moderate" category, traditionally defined as scores between 41% and 60%. Notably, our score is close to the "substantial" category, which begins at 60%, indicating a relatively high level of consensus among the annotators given the complexity of sentiment analysis. The calculated expected agreement (P_e) was 36.01%, which is significantly lower than the observed Kappa score. This indicates that our annotation process is far from random, reflecting a robust and reliable procedure, as κ specifically accounts for the likelihood of chance agreement, which the expected agreement percentage does not. The higher observed κ score relative to the expected agreement underscores the effectiveness of our annotation protocol and the consistency of our annotators' decisions.

In the context of Couple Dialogue data, where mixed emotions, sarcasm, and informal language are prevalent, achieving a κ value close to the substantial category further underscores the robustness of our annotation process. Despite the complexities, our annotators were able to reach a relatively high level of agreement, validating the consistency and reliability of our annotations.

In addition to these metrics, we calculated the average agreement specifically among the four volunteer annotators, which was lower at 72.48% compared to the overall score of 74.02%. This slight reduction in agreement among the volunteers was expected and suggests some variability in their interpretations of the sentiment categories. However, this variability is natural in human annotation tasks and does

TABLE 3. Summary of agreement scores, kappa statistics, and data distribution in the Couple Dialogue dataset. The table reflects the results of annotating 10% of the dialogues (1609 utterances) selected randomly from the dataset.

Metric	Value
Overall Agreement Score	74.02%
Subjectivity/Objectivity Agreement Score	78.06%
Fleiss' Kappa Score (κ)	59.4%
Expected Agreement (P_e)	36.01%
Volunteer Annotators' Agreement Score	72.48%
Data Distribution	Count
Negative Samples	656
Neutral Samples	631
Positive Samples	322

not detract from the overall reliability of our dataset. The high level of agreement, even among volunteers, supports the consistency of the sentiment labels. It is important to note that the labels provided by these annotators were specifically used to assess the reliability of our initial ground truth labels. This validation process was essential to ensure that the ground truth labels were robust and consistent before being used in all experiments. Given the 74.02% agreement achieved with 10% of the data, our baseline annotations, which were considered as ground truth in all experiments involving both Contextual and Non-Contextual Models without requiring further refinement during the volunteer annotation process, can be considered reliable. We have included all the relevant statistics and data distribution in Table 3 for clarity.

IV. METHODOLOGY

In this study, we started with a non-contextual architecture and proceeded with a Contextual Model to predict the sentiment of each utterance in the compiled Couple Dialogue dataset. In this section, we first explain the proposed models and illustrate their architecture in general terms. Then, we explain the preprocessing stage in detail and mention the generated preprocessing modules. Additionally, we provide detailed information about the morphological analyses specific to Turkish.

A. MODELS

1) THE NON-CONTEXTUAL MODEL

Our baseline Non-Contextual Model utilizes embeddings such as BoW, TF-IDF, Word2Vec, and pre-trained Turkish BERT with standard machine learning and deep learning methods. The Non-Contextual Model assumes that all utterances in a conversation are independent of each other. In other words, there is no information flow among the contextual utterances. This type of architecture ignores the speaker's awareness, the context of preceding utterances, and the impact of one speaker on other speakers in the conversation. It treats each utterance as an independent event.

Suppose that our dataset M consists of multiple dialogues, $M = \{D_i\}_{i=1}^N$ where N represents the number of dialogues. Each dialogue involves an input sequence of T utterances $[(u_1, s_1), (u_2, s_2), \dots, (u_T, s_T)]$ where each utterance

$u_i = [u_{i,1}, u_{i,2}, \dots, u_{i,K}]$ encompasses K words $u_{i,j}$ and is uttered by speaker s_i . Our goal is to predict the sentimental category $sent_i$ corresponding to each utterance u_i , where $sent_i \in C$ includes all sentimental categories, including positive, negative, and neutral, utilizing the non-contextual classifier. The plot of our context-independent architecture is given as Fig. 3.

In our approach, each utterance undergoes an initial stage of basic preprocessing. This step is followed by three distinct preprocessing forms: detailed preprocessing applied on the root form, partial preprocessing applied on the root form, and partial preprocessing applied on the surface form, the latter involving morphological analysis. These forms are designed to capture varying levels of linguistic detail, addressing different aspects of language processing, such as the inclusion or exclusion of stopwords, punctuation, and significant morphemes. Feature extraction and subsequent modeling applications are then performed independently for each preprocessing form. The preprocessed forms serve as inputs for both conventional classifiers and deep learning models, including pre-trained Turkish BERT embeddings.

Fig. 3 illustrates the process flow of the Non-Contextual Model. This model begins with basic preprocessing and then applies three different preprocessing forms: detailed preprocessing on the root form, partial preprocessing on the root form, and partial preprocessing on the surface form (involving morphological analysis). Feature extraction techniques, such as BoW, TF-IDF, Word2Vec, and GloVe, are used to create representations of the utterances after these preprocessing steps, which are then fed into conventional machine learning classifiers.

BERT embeddings, derived after basic preprocessing, are applied to both conventional machine learning classifiers and deep learning models. Additionally, GloVe embeddings from basic preprocessing are specifically applied to deep learning models only. This approach inherently disregards contextual dependencies between utterances, focusing solely on the content of individual utterances for sentiment prediction.

2) THE CONTEXTUAL MODEL

In contrast, this model captures the flow of conversation by incorporating speaker-specific context, previous utterances, and the interactions between different speakers, allowing for a more nuanced understanding of sentiment dynamics. Let our dataset M consist of multiple dialogues, $M = \{D_i\}_{i=1}^N$ where N represents the number of dialogues. Each dialogue involves an input sequence of T utterances $[(u_1, s_1), (u_2, s_2), \dots, (u_T, s_T)]$ where each utterance $u_i = [u_{i,1}, u_{i,2}, \dots, u_{i,K}]$ encompasses K words $u_{i,j}$ and is uttered by speaker s_i . Our goal is to predict the sentimental category $sent_i$ corresponding to each utterance u_i , where $sent_i \in C$ with C , being the set of all sentimental categories, including positive, negative, and neutral, by taking into account conversational dynamics utilizing the contextual classifier. Our context-dependent architecture schema is given as Fig. 4.

The Contextual Model, illustrated in Fig. 4, first applies basic preprocessing to the dialogue data, similar to the Non-Contextual Model. However, unlike the Non-Contextual Model, which treats each utterance as an independent event, this model leverages information from previous utterances, speaker states, and emotional dynamics to predict sentiment. Unlike the Non-Contextual Model, the Contextual Model does not include morphological analysis in its preprocessing steps. Instead, it focuses on leveraging the sequential context within the conversation for sentiment prediction. This model is designed to reflect the natural flow of conversation, where each utterance is influenced by those that came before it, as well as by the speaker's ongoing emotional state.

This model utilizes a combination of prompt-based modeling, fine-tuning, and embedding-based approaches to incorporate the context, speaker state, and emotion flow across the dialogue. Prompt creation feeds into both instruction-based and few-shot prompting, with models like GPT-4 and Llama 2 employed for generating prompts that capture the nuances of the conversation. Additionally, fine-tuned models such as Llama 2, GPT-3.5 Turbo, and pre-trained Turkish BERT are used to refine the predictions by integrating speaker-specific contexts and preceding utterances.

The architecture further leverages embedding-based modeling, where DialogueRNN plays a critical role by using extracted embeddings (e.g., from GPT, pre-trained Turkish BERT, fine-tuned Turkish BERT, and GloVe) to model emotional states, global contextual features, and speaker-specific characteristics. This comprehensive approach ensures that each utterance is processed in relation to its context, allowing the model to predict sentiment with a deeper understanding of the dialogue's flow. By maintaining the interdependencies between utterances and the evolving emotional dynamics, the Contextual Model better reflects the intricacies of real-world conversations, thereby improving the accuracy of sentiment predictions.

While the Non-Contextual Model focuses on processing utterances independently after basic preprocessing, the Contextual Model goes beyond capturing the conversational context and dynamics. This distinction is crucial for understanding the strengths and limitations of each approach and is clearly represented in Fig. 3 and 4.

In this study, Fig. 3 and Fig. 4 diagrams were created using draw.io,³ a freely available online diagramming tool.

B. TEXT PREPROCESSING STEPS

Significant portions of natural language processing tasks necessitate data preprocessing techniques as a primary step. This is due to the fact that the effectiveness of machine learning algorithms applied to text is greatly influenced by the preprocessing methods employed. The objective of data preprocessing is to enhance data quality by eliminating noise

³<https://app.diagrams.net/>

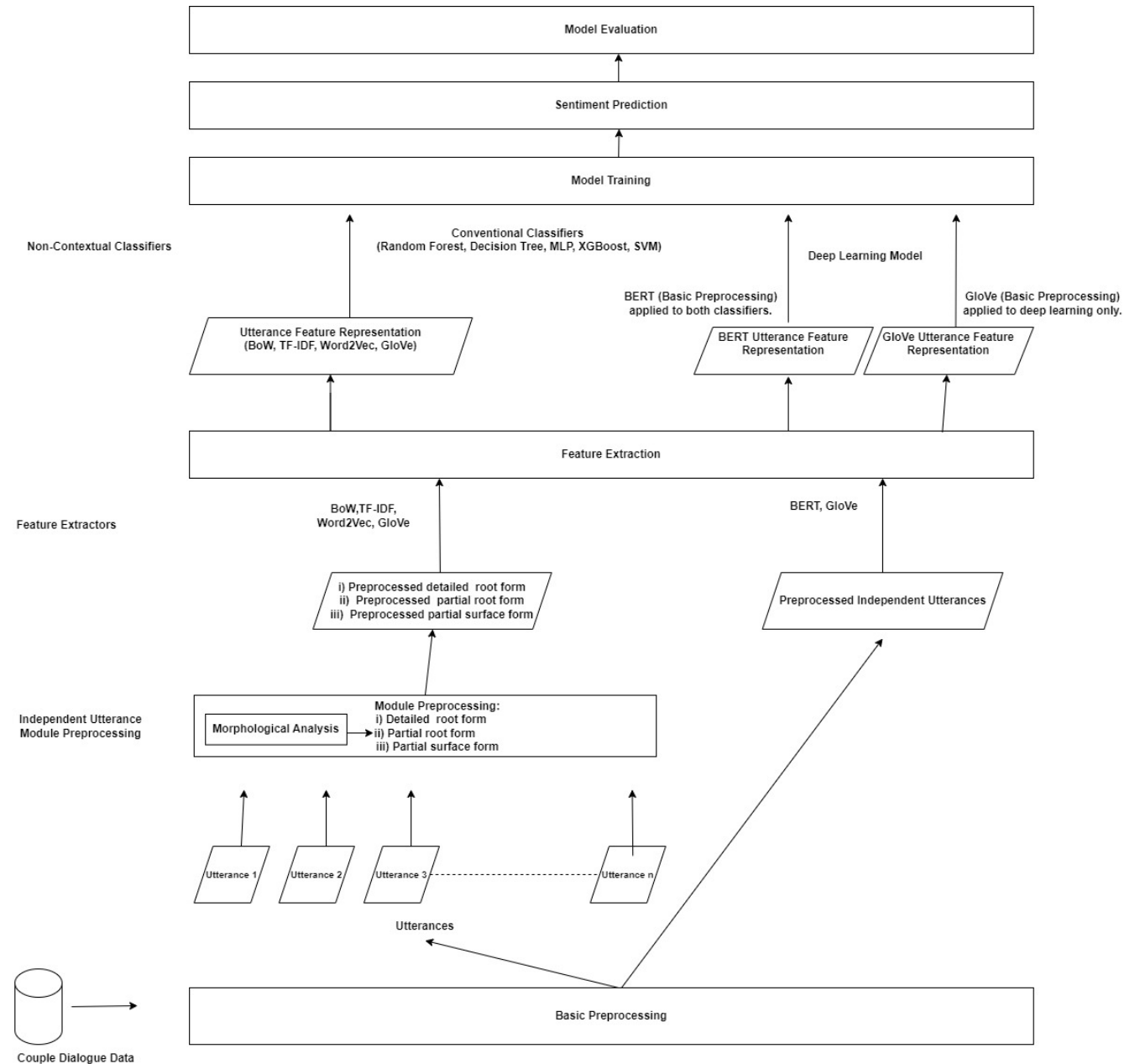


FIGURE 3. Non-contextual sentiment detection flow. Independent utterance processing using feature extractors (BoW, TF-IDF, Word2Vec, GloVe, BERT) for sentiment classification.

through data cleaning techniques, reducing data variability, and standardizing each word.

1) DATA CLEANING

Typically, the data cleansing process involves various tasks, such as converting data into appropriate formats using standard encodings like ISO/IEC or UTF-8, expanding contractions (e.g., ‘we’ll’ to ‘we will’), eliminating undesired elements like HTML/XML marks, punctuation, and numerical values, as well as single characters and excessive spaces. Other steps include generating tokens by segmenting

text into smaller units like words, sentences, or n-grams, removing stop words, converting all text to lowercase for standardization, implementing stemming to reduce terms to their root form, and lemmatization to convert related words to their dictionary format.

We engaged in discussions in Turkish, employing ISO encoding, expanding contractions, eliminating numerical values, single characters, and excessive spaces, and converting the entire text to lowercase for both the Contextual Model and the Non-Contextual Model. This set of preprocessing steps was collectively named “basic-preprocessing.”

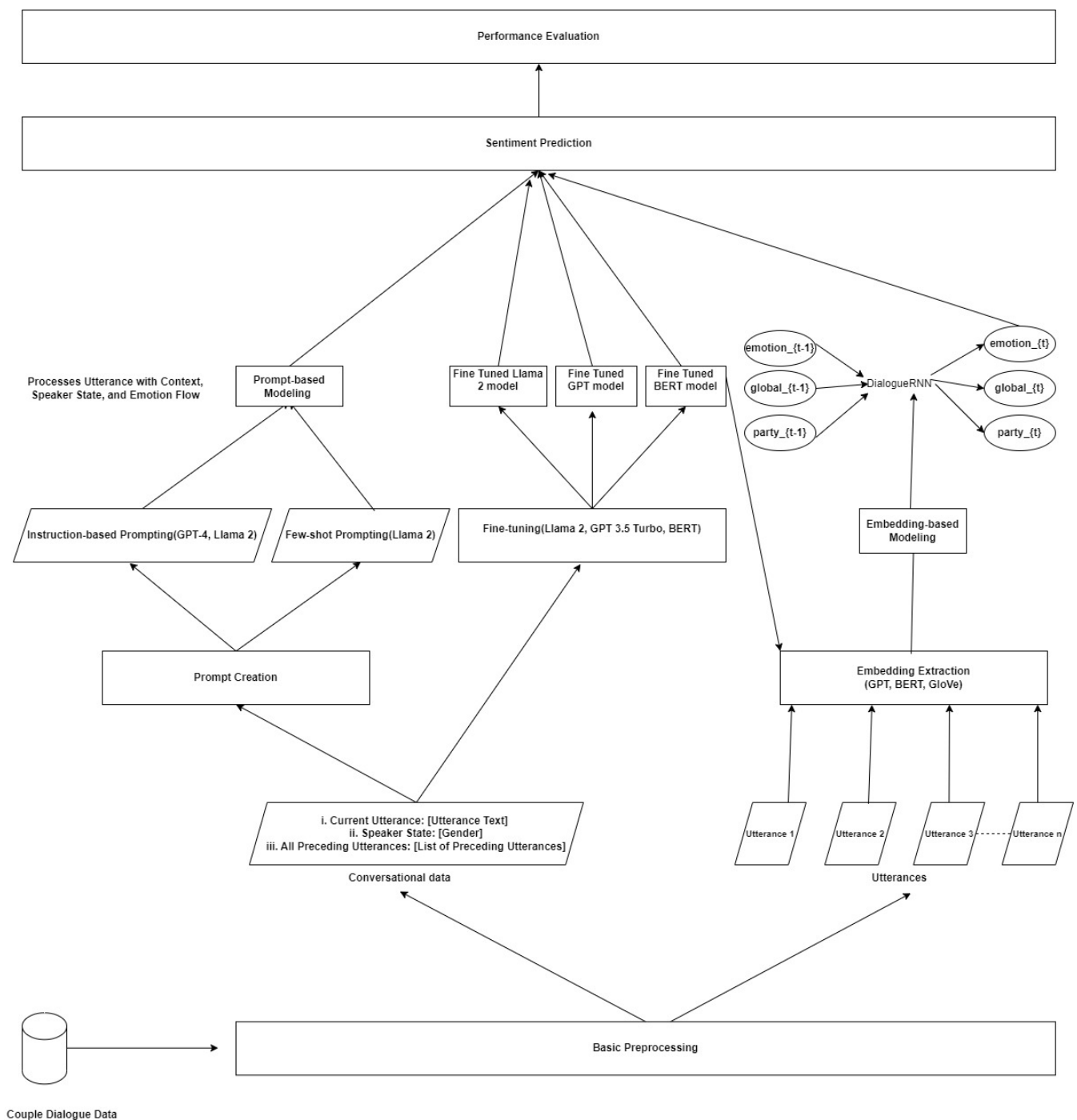


FIGURE 4. Contextual sentiment detection flow. Integrated processing of utterances using conversational dynamics and speaker context for sentiment classification.

2) MORPHOLOGICAL PARSING

Turkish is known for its rich morphological composition, so we aimed to examine the impact of certain suffixes on sentiment classification tasks for our Non-Contextual Model. For this purpose, we utilized a morphological parser and disambiguation tool to obtain the morphological analysis of each word. In this study, we employed a morphological

analyzer called Dilbaz⁴ [90], which facilitates two levels of morphological analysis tasks. Initially, at the word level, we used the morphological analyzer to identify each word, particularly those that were not successfully parsed. Some unrecognizable Turkish words contained misspellings, such

⁴<https://github.com/StarlangSoftware/TurkishMorphologicalAnalysis>

as “çıkacağım” instead of “çıkacağım,” some were written in their English forms, for example, “PlayStation,” while others were compound nouns, like “Heybeliada.” We corrected the misspelled words using the Turkish SpellChecker⁵ tool [91].

Subsequently, we conducted sentence-level morphological analysis, where the morphological disambiguation tool selects the most accurate morphological parse of a word from its potential parses, enabling us to obtain the surface form of each word within the utterance.

The preprocessing steps may vary depending on the structure of the language. Considering that Turkish is characterized by agglutination and possesses a complex morphological structure, removing suffixes may impact performance. Certain suffixes may carry significant sentimental meaning. Therefore, we generated a partial surface form that includes these suffixes if they are present in the surface form of a specified word, as employed by the authors in [67].

We focused on three discerning morphemes: negation morphemes, morphemes conveying a sense of pity, and morphemes expressing wishes, respectively. Negation is a critical feature in sentiment analysis, as it can significantly alter the sentiment of an utterance. Expressions of pity can add a layer of emotional depth to utterances. Wishing forms express hope or desire, which are important in understanding the sentiment of an utterance. We detected negation morphemes within certain verbal suffixes (e.g., *ma/me*) and word morphemes like “*siz*,” “*suz*,” and “*süz*.” For instance, while “*yapım*” (I did) holds a positive connotation, “*yapmadım*” (I did not do) conveys a negative meaning. Similarly, the word “*mide*” (stomach) takes on a “*midesiz*” (tasteless) meaning when it acquires the “*siz*” morpheme.

Another example is “*keşke görse*” (I wish she/he saw), which expresses a desire for something. Finally, “*çocukcagız*” (poor child) conveys a sentiment of pity because of the “*-cagız*” suffix. We observed that certain morphemes are more indicative of the sentiment of words, as depicted in the Results section. The potential preprocessing operations on a sample utterance are presented in Table 4.

3) PREPROCESSING MODULES

For the Non-Contextual Model, we generated three distinct data forms, each tailored to different preprocessing strategies.

Punctuation can lead to data sparsity issues and inflate feature size in word embedding vectors. Additionally, in many cases, stopwords do not hold specific significance for tasks without context. In the first form, we eliminated all punctuation and stopwords, producing the “detailed preprocessing applied on the root form.”

In the second form, which we referred to as the “partial preprocessing applied on the root form,” we did not exclude any stopwords from the content, and we retained exclamation marks and ellipsis punctuations. Some words in the stopword list convey prepositional or negational meaning. Moreover,

stopwords can provide crucial contextual information, especially for contextual models like BERT. Punctuation marks often convey emotional nuances in sentiment analysis tasks. For instance, the exclamation mark “!” in “*Kalk buradan!*” (Get out of here!) intensifies the expression of anger and negative sentiment in the given utterance. Removing such punctuations might strip away essential context, potentially compromising the performance of contextual embeddings.

In the final form, known as “partial preprocessing applied on the surface form,” we extended the second form by labeling a word as “word-partial” if it contained one of the significant suffixes detailed in the Morphological Parsing section. This approach allowed us to capture subtle variations in sentiment conveyed by different morphological forms of the same root word.

Nevertheless, for the Pre-trained Turkish BERT, which is associated with the contextual embedding model, we only applied basic pre-processing tasks, but retained both punctuation marks and stopwords.

V. UTTERANCE FEATURE EXTRACTION

In this research, we utilized a range of word representation techniques to transform text data into vector representations, facilitating their application in various machine learning algorithms. These techniques included “Bag of Words (BoW),” “TF-IDF,” “Word2vec,” “GloVe,” “Pre-trained Turkish BERT,” “Fine-tuned Turkish BERT,” and OpenAI’s “text-embedding-3-small”. For the Non-Contextual Model, we employed BoW, TF-IDF, Word2vec, GloVe, and pre-trained Turkish BERT. The Contextual Model, specifically DialogueRNN, utilized previously extracted embeddings from GloVe, pre-trained Turkish BERT, fine-tuned Turkish BERT, and text-embedding-3-small at the utterance level.

A. BAG OF WORDS (BOW)

BoW [92] is a simple yet effective method to quantify the frequency of words within sentences, transforming them into vectors where the length corresponds to the corpus vocabulary size. We used the Gensim library’s [93] “doc2bow” model to convert text data into BoW vectors. These vectors were then used as input for several machine-learning algorithms to perform sentiment classification.

B. TF-IDF (TERM FREQUENCY-INVERSE DOCUMENT FREQUENCY)

TF-IDF evaluates a word’s relevance to a document in a collection or corpus [94]. It enhances the importance of words unique to specific documents and diminishes the value of words common across all documents. We generated TF-IDF vectors for each utterance and employed them in training classifiers to conduct sentiment analysis effectively.

C. WORD2VEC

Word2vec offers nuanced representations by capturing semantic and syntactic relationships between words [95].

⁵<https://github.com/starlangsoftware/TurkishSpellChecker>

TABLE 4. Examples of preprocessing steps applied to a Turkish sample utterance, showcasing the transformation from raw text to tokenized and morphologically parsed forms, including handling of negation morphemes.

Method	Utterance
Raw text	İçimi boşaltamamış oluyorum, rahatlayamamış oluyorum. (I can't empty myself, I can't relax.)
Tokenization	içimi boşaltamamış oluyorum , rahatlayamamış oluyorum .
Morphological parsing	içimi: iç+noun+A3sg+P1sg+Acc boşaltamamış: boşalt+Verb+ma+mış+Narr+Adj oluyorum: ol+Verb+Pos+Prog1+A1sg ,: ,+Punc rahatlayamamış : rahatla+Verb+ma+mış+Narr+Adj oluyorum: ol+Verb+Pos+Prog1+A1sg ..: ..+Punc
Negation handling	boşaltamamış rahatlamamış

We utilized both the CBOW and Skip-gram models to develop dense vector representations. The setup included vector dimensions of 50, a context window of 3, a minimum word frequency of 2, and employed three worker threads on a corpus comprising 4,656 unique words.

D. GLOVE

GloVe constructs word embeddings using a co-occurrence matrix that captures the frequency of word pairs within a corpus. We applied pre-trained GloVe embeddings with 300 dimensions, trained on the Boun Web Corpus and the Huawei Corpus [96]. These embeddings underwent parameter tuning to optimize embedding quality, including adjustments to the window size and iteration limits. Specifically, the window size was set to 5, the minimum word frequency was set to 10, the maximum number of iterations was limited to 100, and the maximum number of co-occurrences to consider in the weighting function was set at 10.

E. PRE-TRAINED TURKISH BERT

BERT [12] leverages the Transformer architecture [3], which is distinguished by its use of an attention mechanism that comprehensively captures the context of words via bidirectional training. This method, unlike Seq2seq models such as LSTM [2] that rely on sequential data processing, allows for parallel computation, significantly enhancing training efficiency.

The architecture consists of several components: The Embedding Layer utilizes the BertTokenizer for subword tokenization, breaking sentences into manageable pieces.

BertEmbeddings processes each token to generate a word, positional, and type embeddings. These vectors are combined to form a comprehensive sentence embedding, refined through layer normalization and dropout procedures. The Encoder includes 12 layers (BertLayers), each receives inputs from its predecessor. Each layer contains a BertAttention component that calculates attention scores to establish the relevance of words in sentence context, resulting in 12 distinct word representations per layer. The output from the final encoder layer is pooled using the first token, known as the [CLS] token, which is crucial for classification tasks.

In our project focusing on Turkish couple conversations, we employed the Turkish BERT model, specifically pre-trained on a robust dataset of Turkish texts. This model, BERTurk [97], features a 128k vocabulary size and includes 12 encoders with 12 bidirectional self-attention heads each, and a hidden dimension of 768, totaling 110 million parameters. We utilized this configuration to extract utterance-level feature vectors for sentiment analysis within both standard and advanced deep-learning architectures like DialogueRNN. The [CLS] token, extracted from the pre-trained Turkish BERT, was pivotal in both Non-Contextual and Contextual Models for capturing the essence of the utterances effectively, as initially outlined by the creators of BERT.

F. FINE-TUNING TURKISH BERT ON COUPLE DIALOGUE DATA

In this study, we fine-tuned the pre-trained Turkish BERT model on a specially created Couple Dialogue dataset tailored for conversational sentiment analysis. The fine-tuning process was carefully optimized to enhance performance for this

specific task. Key settings included a learning rate of $1e-5$ and a batch size of 16 during both the training and the evaluation phases. The model was trained over three epochs with a weight decay of 0.01 to prevent overfitting. Performance was rigorously monitored by the loss metric, selecting the iteration with the lowest loss for further use.

Once optimal performance was achieved, the model was saved to preserve its state. This enabled the extraction of high-quality embeddings, specifically using the [CLS] token that represents the holistic understanding of the input sequence. These embeddings were then used as inputs for a DialogueRNN architecture, which was designed to perform sentiment analysis within conversational contexts. This approach demonstrates how advanced pre-trained models and fine-tuning techniques can be synergistically used to improve the specificity and accuracy of sentiment classification tasks.

G. TEXT-EMBEDDING-3-SMALL

Our project also incorporated OpenAI's text-embedding-3-small, the latest advancement in text embedding technology. This model has shown significant improvements over its predecessor, text-embedding-ada-002 model,⁶ in terms of both performance metrics and cost efficiency. Notable enhancements were observed in benchmark assessments such as the Multi-Language Retrieval (MIRACL) [98] and the English Task Benchmark (MTEB) [99], where it achieved scores of 44.0% and 62.3%, respectively, marking substantial progress from previous figures. Additionally, the cost-effectiveness of text-embedding-3-small has been dramatically improved, with costs reduced by fivefold from the previous iteration.

Despite the availability of the more powerful text-embedding-3-large model,⁷ we selected text-embedding-3-small for our study due to budgetary considerations. This decision underscores the model's utility in handling complex language tasks effectively within financial constraints. Using the OpenAI Python package and communicating with the model through the OpenAI API, we generated embeddings, each a 1536-dimension vector of floating-point numbers, for every utterance. These embeddings served as crucial feature extractors in the DialogueRNN framework, highlighting the model's robustness in developing AI technologies that require a deep understanding of human language nuances.

H. INCORPORATION OF MORPHOLOGICAL FEATURES

We stated in the Morphological Parsing subsection that we included a morphological parser, Dilbaz, to dissect the intricacies of the Turkish language. The parser effectively handled morphological structures, identifying three key features: negation, pity, and wishing forms, which are crucial for accurately capturing the semantic content of the language [67]. Below, we show how we detected these morphemes.

- **Negation:** We identified negations by seeking the “neg” morpheme in the parsed form.
 - Exp: Yorgun+Adj değil+Neg -im+ Copula (I am not tired)
- **Pity:** We identified pity by seeking the root+Noun+Dim structure.
 - Exp: kız+Noun+Dim (kızcağız - dear/little girl)
- **Wishing:** We identified wishing by seeking the gül+Verb+Cond structure.
 - Exp: gül+Verb+Cond (gülse - if he/she/it laughs)

We underlined the detected forms with the associated morphemes. For instance, we showed gel (come), gelse (if he/she/it comes), and gelmese (if he/she/it does not come) in different forms, treating them as separate entities in the feature space if they are included in the same utterance. As an example, we calculated TF-IDF scores considering gel, gelse, and gelmese differently for the TF-IDF model. Additionally, the co-occurrence score was calculated for the three forms separately for the GloVe embedding model. Likewise, in the BoW model, each morphological form was treated as a distinct token, and in the Word2Vec model, each form was included as a separate word, allowing the model to learn different embeddings for each morphological variant. This separation was only done in the third pre-processing form (“partial preprocessing applied on the surface form”), as morphemes were only regarded in this form.

In contrast, in the second form, referred to as “partial preprocessing applied on the root form,” we did not exclude any stopwords from the content and retained exclamation marks and ellipsis punctuations. Here, gel, gelse, and gelmese were shown with the same vectors since we only used root forms (root: gel). Similarly, in the first form, referred to as “detailed preprocessing applied on the root form,” where we eliminated all punctuation and stopwords, all three were represented as gel, as in the second form.

After creating word embedding models using BoW, TF-IDF, Word2vec, and GloVe, we obtained feature embeddings for the Non-Contextual Model. These embeddings were then used in conventional classifiers under the Non-Contextual Model, and the results were reported in the Experimental Studies and Results section.

VI. UTTERANCE CLASSIFIER

Utterance-level features were extracted using BoW, TF-IDF, Word2Vec, GloVe, and the pre-trained Turkish BERT model. These features were then classified using Non-Contextual Model classifiers, which employ algorithms that disregard the interdependence among different speakers in a conversation.

In contrast, the Contextual Model classifiers, which are designed to capture the flow and dynamics of the conversation, consider both the preceding and subsequent utterances. These classifiers are divided into three categories: prompt-based, fine-tuned, and embedding-based. The embedding-based models utilize a variety of features obtained from GloVe, pre-trained Turkish BERT, fine-tuned

⁶<https://openai.com/blog/new-and-improved-embedding-model>

⁷<https://openai.com/blog/new-embedding-models-and-api-updates>

Turkish BERT, and text-embedding-3-small, enabling a detailed analysis of conversational context.

All computational tasks were executed on a local high-performance workstation equipped with an Intel® Core™ i7-9700K, 64GB of RAM, and an NVIDIA GeForce RTX 4090 GPU with 24 GB of memory.

This section will further elaborate on both the Non-Contextual and Contextual classifiers, explaining their mechanisms and the rationale behind their use.

A. THE NON-CONTEXTUAL CLASSIFIER

We classify the classification methods employed in this study into two categories. The first category involves the application of conventional machine learning algorithms. In the Preprocessing Modules Section, we elaborated on three types of preprocessing modules, namely, “detailed preprocessing applied on the root form”, “partial preprocessing applied on the root form”, and “partial preprocessing applied on the surface form”. These three different data forms of utterance representations obtained from the BoW, TF-IDF, Word2Vec, and GloVe models were employed as inputs for these methods. As BERT constitutes a contextual embedding model, we generated a unified data form of utterance feature representation, implementing fewer preprocessing operations as explained in the Data Cleaning section, and incorporated it into this network.

The second group consists of a deep-learning architecture that utilizes pre-trained GloVe word embeddings and the pre-trained Turkish BERT model as inputs. For these embeddings and the model, we applied basic preprocessing operations to the utterances.

1) CLASSIFICATION ALGORITHMS

The sentiment classification of each utterance is performed using several machine learning algorithms, including Decision Tree, Random Forest, Gradient Boosting, XGBoost, Support Vector Machine (SVM), and Multi-Layer Perceptron. Our generated feature embeddings became the input of those classifiers. All classifiers were implemented using Scikit-Learn library’s algorithms [100].

2) DEEP LEARNING MODEL

Unlike commonly known classifiers, we developed a deep-learning architecture using Keras [101] under the Non-Contextual Model classifier framework. The utterance feature representations of pre-trained GloVe and the pre-trained Turkish BERT model were employed as inputs for this network for the sentiment classification task. It is important to note that these inputs underwent only basic preprocessing before being fed into the model.

We created a BERT layer using Keras and integrated BERT utterance embeddings from TF-Hub for the sentiment classification problem. The BERT layer requires two input layers, namely, input_ids and attention_mask. Keras provides an Embedding layer containing hidden states of the BERT

TABLE 5. Specification of parameters for the deep neural network architecture utilized in the study.

Parameter name	Value
Optimizer	Adam
Epsilon	1e-08
Loss function	Categorical cross-entropy
Maximum length of each sentence	70
Number of epochs	30
Batch size	4

layer. Subsequently, we constructed GlobalMaxPooling1D, Dense, and Dropout layers, and trained our model using the parameters outlined in Table 5. For the GloVe experiment, an Embedding layer encompassed an embedding matrix derived from pre-trained GloVe embeddings.

The constructed architecture including the BERT layer is represented in Fig. 5.

B. THE CONTEXTUAL CLASSIFIER

To provide some background, we explore the advanced contextual language models (LLMs) that have been pivotal in our conversational sentiment analysis task. These models demonstrate significant technological advancements that enhance their suitability for complex linguistic processing.

1) BACKGROUND FOR LLMS

a: BACKGROUND FOR GPT-3.5

GPT-3.5 Turbo acts as a bridge between GPT-3 and GPT-4, maintaining a high parameter count while integrating substantial architectural enhancements. This model improves efficiency in data processing and responsiveness, with optimized attention mechanisms that enhance its performance in dynamic conversational settings. GPT-3.5 Turbo is particularly effective in real-time applications requiring fast and accurate text generation, such as conversational agents.

GPT-3.5 Turbo’s efficiency in processing and generating text makes it ideal for analyzing the rapid and context-rich exchanges typical in real-world couple dialogues. Its ability to quickly grasp and respond to dynamic conversational contexts enhances its effectiveness in sentiment analysis tasks where understanding subtle emotional cues is essential.

b: BACKGROUND FOR GPT-4

Introduced in 2023, GPT-4 extends OpenAI’s legacy with around 175 billion parameters, further enriching the transformer-based architecture. It features an advanced encoder-decoder setup that significantly improves handling longer context windows, crucial for generating coherent, detailed, and contextually relevant outputs. GPT-4’s capabilities make it ideal for complex content generation, extensive dialogue systems, and advanced comprehension tasks, enhancing applications requiring deep linguistic analysis.

GPT-4’s advanced architecture is particularly suited for sentiment analysis in extensive and complex couple dialogues, where maintaining coherence over longer exchanges

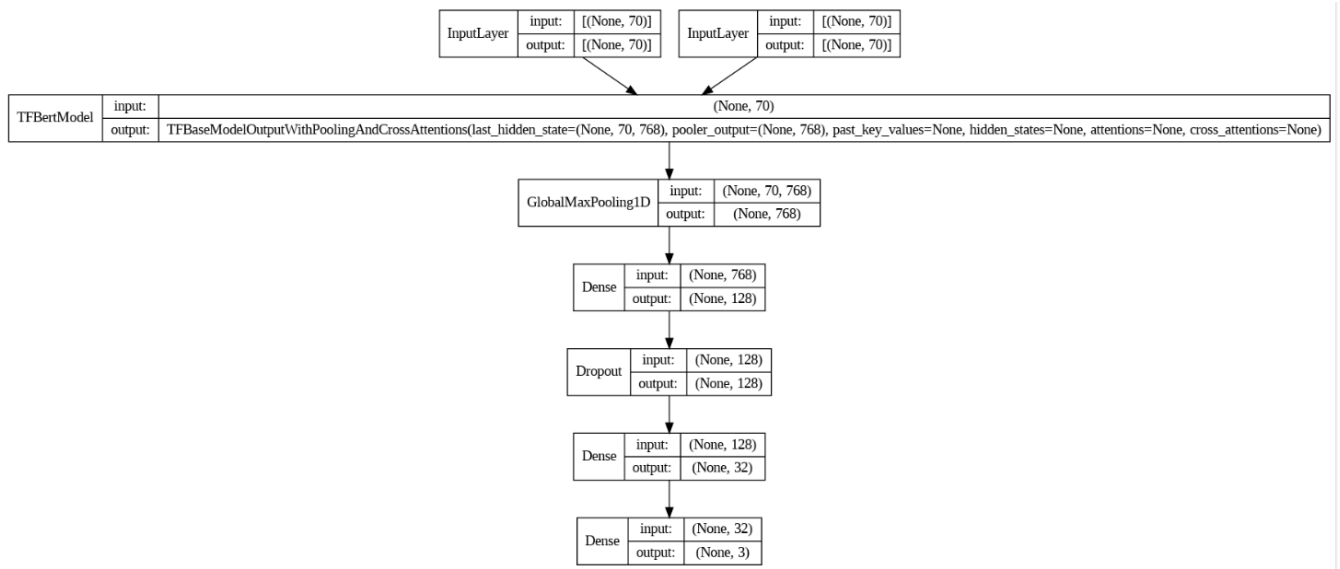


FIGURE 5. Deep learning architecture including a BERT layer.

is crucial. Its ability to handle detailed and contextually relevant outputs ensures accurate identification of sentiment shifts throughout the conversation.

c: BACKGROUND FOR LLAMA 2

Meta GenAI’s LLaMA 2, launched in July 2023 as an open-source initiative, is an advancement from the original LLaMA model, featuring models that range from 7 billion to 70 billion parameters. It supports up to 4K tokens for context length—double that of its predecessor—enabling a deeper understanding of extensive text sequences. This enhancement is crucial for managing intricate dialogues and extended narratives, enhancing the model’s ability to maintain contextual continuity.

Departing from traditional Transformer architectures, LLaMA 2 incorporates Root Mean Square Layer Normalization (RMSQLN) [102] and Swish-Gated Linear Units (SwiGLU) [103] instead of the standard ReLU. These modifications, along with Rotary Position Embedding (RoPE) [104] that merges absolute and relative positional information, enhance the learning dynamics and the attention mechanism. This setup facilitates improved text generation and comprehension, accommodating the positional dynamics of input sequences.

LLaMA 2 undergoes initial training using a self-supervised method on a vast, unlabeled text corpus and is further refined through supervised fine-tuning for specific applications like dialogue systems. It also incorporates reinforcement learning with human feedback (RLHF) [105] to enhance its real-world conversational abilities. For this study, the Trendyol LLM chat variant, based on the LLaMA 2 7B model, was employed for our both prompt-based and fine-tuning approaches.

LLaMA 2’s capability to manage long sequences and maintain contextual integrity is crucial for accurately

analyzing the emotional dynamics in extended couple dialogues. Its refined attention mechanisms and enhanced positional embeddings allow it to effectively process intricate dialogues, making it a strong choice for sentiment analysis in real-world conversational data.

2) PROMPT-BASED MODELING

Prompt engineering is essential in AI applications, as it ensures that prompts are clear and specific, minimizing the likelihood of generating unrelated or overly generalized AI responses. A well-crafted prompt provides the necessary context and directs the model towards producing the desired output style succinctly, making it particularly effective for tasks requiring precise, contextual understanding such as those requiring sentiment analysis.

For our sentiment analysis tasks, we employed two distinct prompting methods with GPT-4 and Llama 2:

- **Instruction-based Prompting:** We designed prompts that directly instruct the AI on its tasks, ensuring responses meet specific, predefined criteria. This approach is implemented across both the GPT-4 and Llama 2 models for our conversational sentiment analysis tasks. This approach is exemplified by a prompt asking the AI to assess the emotional tone of the last conversation in a dialogue as “positive,” “negative,” or “neutral,” and to respond with just one word. Such prompts consider factors like speaker gender and dialogue dynamics, enhancing the accuracy of sentiment assessments.

In crafting these prompts, special attention was given to the nuances of male-female interactions within couple dialogues. This approach ensures that the model can accurately interpret the sentiment based on the specific dynamics and communication patterns present in these

dialogues, which are critical for accurate sentiment assessment.

- **Few-shot Prompting:** This method provides the AI with several “shots” or examples of the task, each comprising an input and the corresponding correct output. This strategy helps the model to recognize the required pattern or task without explicit programming. Our implementation with Llama 2 integrates this approach, comparing its effectiveness against instruction-based results.

The few-shot examples given in this study were carefully selected to represent a range of typical emotional exchanges in male-female interactions within couple dialogues. We prioritized examples that captured various degrees of sentiment intensity and dialogue complexity, ensuring the model could generalize effectively across different types of interactions.

During the development of our prompt structures for conversational sentiment analysis, we conducted numerous trials. We found limited guidance in existing literature, leading us to devise a prompt structure that specifies for each utterance within a dialogue, the AI should be provided with all previous utterances and tasked with predicting the sentiment of the current utterance. Thus, the model is asked to determine the sentiment of the most recent utterance, considering all prior context. Since our dataset is written in Turkish, we applied the instructions in Turkish, as shown in Table 6 and Table 7. The English versions are provided to accommodate readers not familiar with Turkish, ensuring broader accessibility and understanding.

Table 6 illustrates the application of an instruction-based prompt designed for GPT-4 and Llama 2, highlighting the directive nature of our prompts and their focus on critical dialogue aspects to assess sentiment accurately.

Table 7 shows how few-shot prompting can be utilized to enhance the model’s understanding of diverse emotional contexts within dialogues, providing several scenarios with predetermined emotional tones.

By employing these advanced prompt engineering techniques, we have tailored our training, validation, and test datasets to effectively correspond with the required chat completion styles, thus improving the models’ performance in practical AI applications.

a: PROMPT-BASED SENTIMENT PREDICTION WITH GPT-4

In our research, we utilized OpenAI’s GPT-4 for conversational sentiment analysis, emphasizing the critical role of instruction-based prompt engineering. This process involved configuring the technical aspects of the model to enhance its performance for our specific needs. Access to the GPT-4 engine was facilitated through a paid subscription to the OpenAI API, ensuring we could leverage the full capabilities of this advanced tool.

We configured the GPT-4 model as follows: The engine was set to “text-davinci-004” with a temperature of 0.5 to allow a moderate level of creativity in responses. The

response length was limited to 4096 tokens, suitable for longer conversations, and Top_p was set to 1.0, allowing the model to explore all possible response options. No frequency or presence penalties were applied, and responses were clearly segmented by a newline character. These settings enabled us to fine-tune the prompts, maximizing GPT-4’s potential and achieving high accuracy in sentiment analysis.

b: PROMPT-BASED SENTIMENT PREDICTION WITH LLAMA 2

For the practical application of Llama 2, the utilized Trendyol LLM model is built on the open-source Llama 2 framework. This open-source availability makes it accessible for researchers and the general public without the costs associated with commercial platforms like OpenAI. Such accessibility proves invaluable in academic settings, where budget constraints can often limit the scope of research possibilities.

We specifically tailored the configuration of the Trendyol LLM model to meet the requirements of our project. Our custom settings included do_sample=True for generating stochastic responses, a temperature setting of 0.3 to ensure response stability, and selection controls top_k=50 and top_p=0.9 to optimize the relevancy of responses. While the model’s website lists the maximum token size as 1024, we customized our setup to extend this limit to 4096 tokens. This adjustment was necessary to capture all contextual nuances in the conversations we analyzed. These settings, including the extended token size, were specifically chosen to ensure they were perfectly aligned with our conversational sentiment analysis tasks.

The structure of the prompts is particularly crucial for the performance of Llama 2. Our initial instruction-based approach, which directed the AI to label sentiments with minimal context, proved insufficient for achieving the desired accuracy. This led us to implement a few-shot prompting strategy, where providing examples significantly improved the model’s comprehension. By including examples of dialogues depicting various emotional tones as represented in Table 7, Llama 2 was better able to understand and classify sentiments effectively. This adaptability underscores the importance of sophisticated prompt engineering in enhancing the capabilities of AI models for specific analytical tasks.

3) FINE-TUNING

Fine-tuning in machine learning involves adapting a pre-trained model to enhance its performance for a specific task. This technique is especially valuable in natural language processing (NLP), where large, pre-trained models like those from OpenAI can be customized to specific tasks using relatively small datasets. By fine-tuning, we leverage the model’s broad language understanding abilities while optimizing it to capture the specific nuances of a targeted application, such as sentiment analysis within conversational contexts.

TABLE 6. Example of an instruction-based prompt for sentiment analysis tasks using GPT-4 and Llama 2, presented in both Turkish and English. The primary task is conducted in Turkish, with the English translation provided for clarity and broader accessibility.

Turkish Version	English Version
Son konuşmanın duygusal tonunu sadece “pozitif”, “negatif”, ya da “nötr” olarak değerlendir. Cevabın tek bir kelime olmasına dikkat et. Konuşmacı cinsiyeti ve diyalog dinamiklerinin önemini göz önünde bulundur. Sen yardımcı bir asistansın. En iyi cevabı vermek için verilen talimatları dikkate al. Her konuşma kadın veya erkek konuşmacılardan biri tarafından yapılır. “None” öncesi konuşmaların olmadığını gösterir. Yeni Diyalog: Diyalog: Konuşma 1: [Kadın: Gerçekten sinirliyim, sen başlayabilirsin.] Konuşma 2: [Erkek: Ne başlayacağım?] Konuşma 3: [Kadın: Hıh!] Konuşma 4: [Erkek: Hani ne sinirleniyorsun?] Konuşma 5: [Kadın: Arkadaşlar konusuna sen direkt başlayabilirsin bence.] En Sonki Konuşmanın Beklenen Duygu Tonu:	Evaluate the emotional tone of the last conversation as either “positive”, “negative”, or “neutral”. Ensure your response is a single word. Consider the importance of speaker gender and the dynamics of the dialogue. You are a helpful assistant. Make sure your response follows the given instructions. Each conversation is conducted by either a male or female speaker. “None” indicates there are no prior conversations. New Dialogue: Dialogue: Conversation 1: [Female: I’m really annoyed, you start.] Conversation 2: [Male: Start what?] Conversation 3: [Female: Hmph!] Conversation 4: [Male: What are you annoyed about?] Conversation 5: [Female: I think you should start talking about friends directly.] Expected Emotional Tone of the Last Conversation:

TABLE 7. Examples of few-shot prompting for sentiment analysis in dialogues with Llama 2, presented in both Turkish and English. The primary task utilizes the Turkish version, with English translations provided for accessibility and clarity.

Turkish Version	English Version
Görev: Son konuşmanın duygusal tonunu sadece “pozitif”, “negatif”, ya da “nötr” olarak değerlendir. Cevabın tek bir kelime olmasına dikkat et. Konuşmacı cinsiyeti ve diyalog dinamiklerinin önemini göz önünde bulundur. Sen yardımcı bir asistansın. En iyi cevabı vermek için verilen talimatları dikkate al. Her konuşma kadın veya erkek konuşmacılardan biri tarafından yapılır. “None” öncesi konuşmaların olmadığını gösterir.	Task: Evaluate the emotional tone of the last conversation as “positive”, “negative”, or “neutral”. Ensure your response is a single word. Consider the importance of speaker gender and the dynamics of the dialogue. You are a helpful assistant. Make sure your response follows the given instructions. Each conversation is conducted by either a male or female speaker. “None” indicates there are no prior conversations.
Örnek 1 Diyalog: Konuşma 1: [Kadın: Bugün çok güzel bir gün geçirdik.] Konuşma 2: [Erkek: Evet, harikaydı. Seninle olmak her zaman çok güzel.] En Sonki Konuşmanın Beklenen Duygu Tonu: pozitif	Example 1 Dialogue: Conversation 1: [Female: We had a really nice day today.] Conversation 2: [Male: Yes, it was wonderful. Being with you is always great.] Expected Emotional Tone of the Last Conversation: positive
Örnek 2 Diyalog: Konuşma 1: [Erkek: Dün akşamki tartışma için üzgünüm.] Konuşma 2: [Kadın: Ben de. Bir daha olmasın.] En Sonki Konuşmanın Beklenen Duygu Tonu: nötr	Example 2 Dialogue: Conversation 1: [Male: I’m sorry about the argument last night.] Conversation 2: [Female: Me too. Let’s not let it happen again.] Expected Emotional Tone of the Last Conversation: neutral
Örnek 3 Diyalog: Konuşma 1: [Kadın: Seni hiç anlamıyorum!] Konuşma 2: [Erkek: Bunu nasıl söyleyebilirsin?] En Sonki Konuşmanın Beklenen Duygu Tonu: negatif	Example 3 Dialogue: Conversation 1: [Female: I just don’t understand you!] Conversation 2: [Male: How can you say that?] Expected Emotional Tone of the Last Conversation: negative
Yeni Diyalog **Diyalog**: Konuşma 1: [Kadın: Gerçekten sinirliyim, sen başlayabilirsin.] Konuşma 2: [Erkek: Ne başlayacağım?] Konuşma 3: [Kadın: Hıh!] Konuşma 4: [Erkek: Hani ne sinirleniyorsun?] Konuşma 5: [Kadın: Arkadaşlar konusuna sen direkt başlayabilirsin bence.] En Sonki Konuşmanın Beklenen Duygu Tonu:	**New Dialogue** **Dialogue**: Conversation 1: [Female: I’m really annoyed, you start.] Conversation 2: [Male: Start what?] Conversation 3: [Female: Hmph!] Conversation 4: [Male: What are you annoyed about?] Conversation 5: [Female: I think you should start talking about friends directly.] Expected Emotional Tone of the Last Conversation:

a: FINE-TUNING WITH GPT-3.5 TURBO

Due to budget constraints, our research utilized the paid fine-tuning capabilities of the GPT-3.5 Turbo model, specifically the “gpt-3.5-turbo-0125 variant,”⁸ available through the OpenAI API.

1. Fine-Tuning Process:

We structured our dataset, for training, validation, and testing, into “.jsonl” files. Each data entry followed a structured

format where sequences of messages were assigned roles like “system,” “user,” and “assistant.” Table 8 is an example of the data structure used:

In this setup, “new_dialogue” encompasses both the current and all preceding utterances, while “expected_sentiment” classifies the sentiment of the latest utterance as “positive”, “negative”, or “neutral”. The fine-tuning process employs a system prompt meticulously crafted to impartially evaluate the emotional tone of the final conversation. This prompt generates a single-word sentiment, taking into account

⁸<https://platform.openai.com/docs/models/gpt-3-5-turbo>

TABLE 8. A typical JSON format used to represent dialogue data, detailing roles and their corresponding content keys and values for a conversational system.

Role	Content Key	Content Value
system	content	system_prompt
user	content	new_dialogue
assistant	content	expected_sentiment

variables such as speaker gender and dialogue dynamics, similar to the instructional guidance used in our prompt-based strategies for conversational sentiment analysis.

2. Training Strategy:

For the training and validation phases, we provided the model with the label for the last utterance, allowing it to learn the relationship between dialogue context, speaker states (e.g., female, male), and sentiment. This method helped the model understand the intricate dynamics of conversational sentiment. The simplified example is shown in Table 9.

The model sees the entire dialogue and learns that the sentiment of the final utterance (“Kadın: Arkadaşlar konusuna sen direkt başlayabilirsin bence.” / “Female: I think you should start talking about friends directly.”) is negative. The fine-tuning process employs such examples to train the model effectively in evaluating the emotional tone of conversations.

During the fine-tuning process, we initially used OpenAI’s default training parameters, which were as follows: 2 epochs, a learning rate multiplier of 2, and a batch size of 12. These defaults are typically recommended for fine-tuning as they are tailored to the size of the dataset being used. However, as we monitored the training, validation, and total loss metrics via the OpenAI Fine-tuning API, we observed that the validation loss did not decrease significantly after 2 epochs.

To address this, we increased the number of epochs to 3, which led to a more effective reduction in the loss. This adjustment was critical in preventing the model from underfitting and ensuring it could better capture the nuanced patterns in the training data. The validation set was continuously used to monitor performance and ensure the model did not overfit.

With the default parameters and the increased epoch count, the fine-tuned model processed a total of 6,682,384 tokens, respecting the 4096-token capacity of the GPT-3.5 Turbo model.

3. Outcome and Usage:

The fine-tuning session was completed in just 0.78 hours (47 minutes). This fine-tuned model demonstrated effectiveness comparable to that of the DialogueRNN combined with fine-tuned BERT embeddings as shown in the Experimental Studies and Results section, showcasing GPT-3.5 Turbo’s robust capabilities in nuanced conversational sentiment analysis. These results support previous findings that GPT models, through fine-tuning, are adept at handling tasks requiring sophisticated language understanding and generation in dynamic conversational scenarios.

b: FINE-TUNING WITH LLAMA 2

PEFT [106] allows for more efficient fine-tuning of large pre-trained models by updating only a small subset of the model’s parameters, rather than the entire model. This approach preserves the extensive knowledge base the model acquired during its initial training while allowing it to be refined for specific tasks. LoRA [107] specifically introduces low-rank matrices into the transformer layers, enabling targeted adjustments with minimal additional parameters. This technique is particularly useful in maintaining the model’s efficiency while enhancing its performance for specialized tasks.

1. Fine-Tuning Process:

Given our resource constraints, we implemented 4-bit quantization, which significantly reduces the memory requirements of our models. Quantization compresses the model, enabling it to operate within the 24 GB memory limit of our hardware while speeding up inference—an essential feature for real-time applications.

For this project’s conversational sentiment analysis, we fine-tuned the Trendyol LLM, a model based on Llama 2, to optimize it for chatbot interactions. We used the same fine-tuning parameters as those provided by Trendyol LLM’s documentation, with one key modification: the learning rate. While the Trendyol LLM documentation suggests a learning rate of $2e-4$ in the model configuration, we experimented with three smaller learning rates, each lower than $2e-4$, to find the optimal setting. Through this experimentation, we observed that the validation loss decreased most effectively with a learning rate of $1e-6$. Based on these results, we selected $1e-6$ as the final learning rate for fine-tuning, alongside 3 epochs, to ensure a more gradual adaptation of the model. This combination helped reduce the risk of overfitting while allowing the model to learn effectively from the smaller and more specialized dataset used in this project.

The other parameters were set as follows:

- **LoRA alpha:** 128, which controls the scaling of the low-rank matrices introduced by LoRA.
- **Dropout rate:** 0.05, to prevent overfitting by randomly dropping units during training.
- **LoRA rank:** 64, specifying the rank of the low-rank matrices, focusing on all linear modules involved in causal language modeling.
- **Compute dtype:** bfloat16, chosen for its efficiency in handling the 4-bit quantized model.

2. Training Strategy:

Training was conducted over three epochs with a micro-batch size of two, using an AdamW optimizer with torch-fused operations. This optimizer was selected due to its effectiveness in handling the nuances of transformer-based architectures. The maximum sequence length was set to 4096 tokens to ensure the model retained comprehensive context from all previous utterances, which is crucial for understanding the flow of dialogue.

TABLE 9. Example of a JSON structure used in the fine-tuning process for sentiment analysis, presented in both Turkish and English. The table illustrates how dialogue data is structured, including roles such as “system,” “user,” and “assistant,” to help the model learn the relationship between dialogue context, speaker states, and sentiment.

Turkish Version	English Version
"system" : {"content": "Son konuşmanın duygusal tonunu sadece “pozitif”, “negatif”, ya da “nötr” olarak değerlendir. Cevabın tek bir kelime olmasına dikkat et. Konuşmacı cinsiyeti ve diyalog dinamiklerinin önemini göz önünde bulundur."}, "user" : {"content": "[Kadın: Gerçekten sinirliyim, sen başlayabilirsin.] [Erkek: Ne başlayacağım?] [Kadın: Hıh!] [Erkek: Hani ne sinirleniyorsun?] [Kadın: Arkadaşlar konusuna sen direkt başlayabilirsin bence.]"}, "assistant" : {"content": "negatif"}	"system" : {"content": "Evaluate the emotional tone of the last conversation as “positive”, “negative”, or “neutral”. Ensure your response is a single word. Consider the importance of speaker gender and the dynamics of the dialogue."}, "user" : {"content": "[Female: I’m really annoyed, you can start.] [Male: Start what?] [Female: Hmph!] [Male: What are you annoyed about?] [Female: I think you should start talking about friends directly.]"}, "assistant" : {"content": "negative"}

TABLE 10. Detailed specifications of the parameters used in the fine-tuning process of the Trendyol LLM, which is built upon the Llama 2 framework. These parameters were customized for optimizing performance on the Couple Dialogue dataset used in this study.

Parameter	Value
Quantization Type	4-bit ('nf4' double quantization)
Compute dtype	bfloat16
LoRA Alpha	128
Dropout Rate	0.05
LoRA Rank	64
Optimizer	AdamW with torch fused operations
Learning Rate	1e-6 (constant)
Epochs	3
Micro Batch Size	2
Maximum Sequence Length	4096 tokens
Evaluation Frequency	Every 25 steps
Total Training Duration	47 hours

Since we obtained significant improvement in Weighted F1 scores with Few-shot Prompting compared to Instruction-based Prompting, we fine-tuned our data, including few-shot examples similar to sentiment examples given in Table 7. During the training and validation phases, the model was provided with labels for the current utterance, enabling it to learn the relationships between dialogue context, speaker states (e.g., female, male), and sentiment. In contrast, during the testing phase, the model was required to predict the sentiment without being shown the label, assessing its ability to generalize from learned examples.

It is important to note that, due to budget constraints, we were unable to incorporate such extensive examples into the fine-tuning of GPT-3.5 Turbo.

3. Outcome and Usage:

Evaluations were conducted every 25 steps to continuously refine the model’s performance, with the entire process lasting 47 hours. This training regime aimed to achieve minimal loss and was meticulously tracked using TensorBoard for detailed analysis, aligning with best practices in recent research. The overall parameter list is represented in Table 10.

c: FINE-TUNING WITH PRE-TRAINED TURKISH BERT

In this study, we fine-tuned the pre-trained Turkish BERT model on a specialized Couple Dialogue dataset tailored for conversational sentiment analysis. The fine-tuning process

was carefully optimized to enhance performance for this specific task.

1. Fine-Tuning Process:

We utilized a learning rate of 1e-5 and a batch size of 16 during both the training and evaluation phases. These settings were chosen based on initial testing to balance training speed and model performance. The model was trained over three epochs, with the number of epochs determined based on the validation loss to ensure optimal performance without overfitting.

A weight decay of 0.01 was applied to further prevent overfitting while allowing the model to learn effectively from the data. Performance was rigorously monitored using the loss metric, and the model corresponding to the iteration with the lowest loss was saved. This approach ensured that the best-performing model was preserved for subsequent use in sentiment analysis.

2. Training Strategy:

During fine-tuning, all preceding utterances of the current dialogue, the current utterance label, and speaker states (e.g., female, male) were included in the training and validation phases. This comprehensive inclusion allowed the model to learn the relationships between dialogue context, speaker characteristics, and sentiment. In the testing phase, the model was evaluated on its ability to predict sentiment labels without access to the correct answers, thus assessing its generalization capability.

3. Outcome and Usage:

- **Embedding Extraction:** The saved model was used to extract high-quality embeddings from the [CLS] token, which captures the holistic essence of the dialogues. These fine-tuned Turkish BERT embeddings were then utilized within a DialogueRNN architecture for sentiment analysis, as explained in the Utterance Feature Extraction section.
- **Direct Sentiment Classification:** The fine-tuned model directly classified sentiments within these conversations, demonstrating its precision in sentiment detection and its broad applicability in sentiment analysis tasks.

4) EMBEDDING-BASED MODELING

This section details the technical implementation of DialogueRNN, which utilizes a variety of embedding sources

to enhance its dialogue processing capabilities. Specifically, the model incorporates embeddings from pre-trained Turkish BERT, its fine-tuned variant, and what we refer to as text-embedding-3-small, herein identified as GPT-embeddings. Additionally, GloVe embeddings are used, all of which are discussed in the Utterance Feature Extraction section.

DialogueRNN Approach:

DialogueRNN [29] effectively models the dynamics of conversation by tracking changes in the emotional state of each speaker, the evolving context of the dialogue, and interactions across utterances. It achieves this through a combination of party, global, and emotional state representations, updated via gated recurrent units (GRU) [108].

The model enhances traditional seq2seq approaches by incorporating attention mechanisms, as detailed in prior studies [109], [110], to focus on relevant parts of the input sequence dynamically. This attention is particularly crucial when processing extensive dialogues, aiding the model in maintaining a consistent understanding of the context throughout the conversation. DialogueRNN model considers three states:

- **Global State:** Utilizes past utterances and states to model the current context of a dialogue. The global state is updated by a GRU and refined through an attention mechanism to ensure relevance to the current utterance.
- **Party State:** Represents individual speaker states throughout the dialogue, updated with each new utterance to reflect the latest conversational context and content.
- **Emotion/Sentiment State:** Initially designed to track emotions, this has been adapted in our study to the sentiment state to classify sentiments as positive, negative, or neutral effectively. This state is crucial for our conversational sentiment analysis, determining the sentiment based on the current dialogue dynamics.

Training and Evaluation:

We utilized both k-fold cross-validation and standard split methods to evaluate the models. K-fold cross-validation provides a more comprehensive assessment by ensuring that every data point is used for both training and validation, which is especially useful in smaller datasets. In contrast, the standard split method was chosen for certain models due to its lower computational requirements, making it more feasible for models with extensive processing demands, such as those utilizing LLMs.

a: K-FOLD CROSS-VALIDATION

A 10-fold cross-validation was conducted for the DialogueRNN classifier, where each fold consisted of 118 couple dialogues split into 7 training, 1 validation, and 2 testing folds. We optimized hyperparameters, specifically learning rate and epoch size, during the validation phase.

b: STANDARD SPLIT TRAINING

For our experiments, we implemented a dataset split of 70% for training, 10% for validation, and 20% for testing, applied

TABLE 11. Detailed Hyperparameter Settings for DialogueRNN variants: The hyperparameters for different embedding technologies—GloVe, BERT Variants, and GPT—used within the DialogueRNN framework.

Parameter	GloVe	BERT-Variants	GPT
Optimizer	AdamW	AdamW	AdamW
Loss Function	Categorical Cross-Entropy		
Batch Size	1	1	1
Output Size (D_G, D_P, D_E)	100	200	200

TABLE 12. Training times for different embedding types for DialogueRNN architecture.

Embedding Type	K-Fold Validation Time	Cross Standard Split Time
Pre-trained Turkish BERT	93 hours	17 hours
Fine-tuned Turkish BERT	41 hours	9 hours
Text-Embedding-3-Small	43 hours	6 hours
GloVe	47 hours	7.5 hours

uniformly to both the computationally demanding LLMs and the DialogueRNN model. This consistent evaluation setup ensured direct comparisons could be made between the DialogueRNN and the more computationally advanced LLMs.

We employed k-fold cross-validation for the Non-Contextual and DialogueRNN classifiers to thoroughly test using each part of the dataset. In contrast, LLM-based classifiers were evaluated using a standard split, aligning with common practices for large models.

c: HYPERPARAMETERS AND MODEL CONFIGURATION

The standard split hyperparameters were:

- **Glove-DialogueRNN:** Learning rate of 1×10^{-3} , 30 epochs.
- **Bert-DialogueRNN and Other Bert Variants:** Learning rate of 1×10^{-5} , 50 epochs.
- **GPT-DialogueRNN:** Learning rate of 1×10^{-4} , 40 epochs.

Other model parameters for the k-fold cross-validation and standard split are given in Table 11. All models were trained with a batch size of one dialogue, due to memory constraints. The optimizer, loss function, and output size dimensions (D_G, D_P, D_E) for all Bert variants, GloVe, and GPT-embedding were consistently maintained as per foundational parameters outlined in the literature [48].

Our approach in the Embedding-based Modeling part included extracting embeddings and integrating them into the DialogueRNN framework, followed by hyperparameter optimization for learning rate and epoch numbers during the validation phases, and evaluating the results on test data. Table 12 displays the training durations for each embedding type under both k-fold and standard split setups. As anticipated, the k-fold cross-validation required longer training times due to repeated hyperparameter tuning.

The training times for embeddings extracted from the pre-trained Turkish BERT were longer than those from the fine-tuned BERT model because the pre-trained model required additional steps to adapt the general-purpose embeddings to the specific nuances of the Couple Dialogue dataset. Furthermore, these times include both the extraction of embeddings and the subsequent processing with the DialogueRNN architecture.

C. HYPERPARAMETER TUNING

This study employed a 10-fold cross-validation method to stratify the data for both Non-Contextual and DialogueRNN classifiers within the Contextual Model category. Each fold contained 118 couple dialogues, segmented into 7 training, 1 validation, and 2 testing folds. Due to the extensive computational demands of LLMs, k-fold cross-validation was deemed impractical for these classifiers. Instead, the dataset was divided into segments of 70% for training, 10% for validation, and 20% for testing to ensure uniform evaluation conditions across all models.

Hyperparameter optimization was conducted using the Optuna framework [111], aiming to maximize the Weighted-average F1 score. Optuna was selected for hyperparameter optimization due to its ability to efficiently explore a large search space using techniques like Bayesian optimization. This made it particularly effective in identifying optimal parameters with fewer iterations, saving both time and computational resources compared to traditional grid search methods. The optimization process involved conducting 50 trials to determine the best parameters, which were then applied to train models on the training folds and evaluate them on the testing folds. This procedure was iterated for each validation fold, with the average performance across all folds reported.

The parameters tuned included:

- **Tree-based algorithms** (Decision Tree, Random Forest, Gradient Boosting, and XGBoost): tree depth, number of estimators, and learning rate.
- **Support Vector Machine (SVM)**: C value, gamma, and kernel type.
- **Multi-Layer Perceptron (MLP)**: activation function, alpha value, hidden layer sizes, learning rate, and solver type.

For the Deep Learning Model under our Non-Contextual Model, hyperparameter optimization was also performed. For each validation fold of the 10 folds, we fine-tuned the dropout value, activation function, and learning rate. Dropout was adjusted to prevent overfitting, while the activation function and learning rate were tuned to balance model convergence speed and accuracy. The optimal hyperparameters were then applied to the test folds, and the average weighted F1 score was reported, similar to the other Non-Contextual classifiers.

As discussed in the Embedding-based Modeling section, DialogueRNN also underwent hyperparameter optimization,

specifically focusing on learning rate and epoch number adjustments during the validation phase of the k-fold cross-validation and standard split training setups.

D. EVALUATION METRICS

Due to the imbalanced nature of our dataset, we adopted the Weighted-averaged F1 score to assess the performance of our classifiers. This metric was crucial for representing our success rates accurately, especially given the distribution of classes as depicted in Fig. 1, where only 2083 out of 14294 utterances are positive. The Weighted F1 score is particularly effective in unbalanced datasets as it considers both precision and recall, crucial for a comprehensive evaluation of model performance.

$$\text{recall} = \frac{TP}{TP + FN} \quad (2)$$

$$\text{precision} = \frac{TP}{TP + FP} \quad (3)$$

$$F_1 = 2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}} \quad (4)$$

Given the skewed distribution of our dataset, focusing solely on per-class scores could lead to misleading interpretations of model efficacy. Therefore, the Weighted-averaged F1 score is computed by averaging the F1 scores of each class, weighted by the number of true instances per class (i.e., the support of each class). This approach ensures that classes with more instances disproportionately influence the overall score, aligning the evaluation with the actual prevalence of each class. In practice, if the positive class represents only 15% of the data, the Weighted-averaged F1 score will adjust its contribution accordingly, ensuring that the overall metric reflects the true distribution of classes. This method provides a more balanced and realistic assessment of classifier performance across the diverse conditions presented by our dataset.

The complete pseudocode of the conversational sentiment analysis on the Couple Dialogue dataset is provided in Appendix C.

VII. EXPERIMENTAL STUDIES AND RESULTS

A. NON-CONTEXTUAL CLASSIFIER RESULTS

1) CONVENTIONAL CLASSIFIER RESULTS

The tables in this section display the average Weighted F1 scores, derived from two testing folds within a 10-fold cross-validation framework. Each cross-validation cycle involved segmenting 118 couple dialogues into seven training, one validation, and two testing folds. The results reported here represent the average of the Weighted F1 scores across these testing folds.

Table 13, Table 14, Table 15, and Table 16 present the classification performances on data prepared by three different types of preprocessing modules when the utterances are considered independent of each other.

a: BoW

Table 13 displays the Weighted F1 scores of the BoW when we applied several machine learning algorithms on three different data forms. We observe a general trend of improvement in performance when moving from the Root form + Detailed Preprocessing data form to Partial surface form + Partial Preprocessing for the BoW model feature extractor.

As couples used informal language to address issues about their marriage, the inclusion of stopwords and punctuation, including exclamation marks and ellipses, contributed to the sentiment classification task. Each classifier exhibited a notable positive performance impact upon incorporating stopwords and specified punctuations. Furthermore, flagging a word when it includes one of the morphemes carrying pity, sentimental meaning, wishes, or negation morphemes facilitated the utterance label classification task, as observed in the Partial surface form + Partial Preprocessing row in Table 13 in general.

b: TF-IDF

On the other hand, the inclusion of certain morpheme forms did not always yield a positive impact on TF-IDF word embeddings, as is evident in Table 14, in terms of Weighted F1 performance. Nevertheless, it is noteworthy that the incorporation of stop words and specific punctuations consistently demonstrated a significant improvement in classification scores for the utterance features extracted by the TF-IDF model.

c: Word2Vec

Word2Vec is recognized for its ability to understand and represent the semantic relationships between words. This neural network-based technique significantly surpasses the limitations of both BoW and TF-IDF by capitalizing on contextual information. TF-IDF addresses BoW's drawback of treating all words equally by assigning higher weights to less common words, thereby emphasizing their potential importance in text analysis. Improvements in classification performance are not consistently seen across classifiers when transitioning from traditional methods like BoW to more advanced embeddings such as Word2Vec. Although Word2Vec offers significant semantic enhancements, it does not always surpass the performance of simpler models. For example, as indicated in Table 13, traditional BoW features sometimes outperform both TF-IDF and Word2Vec in setups involving SVM (54.63%), XGBoost (55.01%), and Multi-Layer Perceptron (53.32%).

Remarkably, XGBoost consistently benefits from Word2Vec embeddings, as evidenced by an F1 score of 54.31%. This suggests that XGBoost is particularly adept at utilizing the semantic depth offered by Word2Vec, likely due to its capability to manage complex data interactions more effectively. The preprocessing approaches—ranging from detailed to partial processing, which includes the preservation

of contextual nuances through morphemes—may further enhance this performance, enabling XGBoost to leverage these nuanced semantic signals optimally.

The classification effectiveness across all models generally increases with less stringent preprocessing, as detailed in Table 15. For example, the Decision Tree classifier's F1 score rises from 45.53% under detailed preprocessing to 51.04% with partial preprocessing in root forms, though it slightly drops to 49.91% when partial surface forms are used. This trend underscores the potential of moderate preprocessing to enhance generalizability and feature capture by classifiers.

d: GloVe

While the GloVe model is based on analyzing word-to-word co-occurrence statistics from the entire corpus, Word2Vec focuses on co-occurrence within the local context. We expected the pre-trained GloVe embeddings to yield better Weighted F1 scores due to its more advanced modeling approach. However, when we applied the same classification algorithms to the utterance features generated with these embeddings, the results were similar to those obtained with Word2Vec, contrary to our expectations, as shown in Table 16.

e: A PAIRED T-TEST STUDY OF MORPHEME EFFECTS ON SENTIMENT ANALYSIS

The analysis presented in Table 13, Table 14, Table 15, and Table 16 illustrates a consistent trend of performance enhancement from the Root form + Detailed Preprocessing to Partial surface form + Partial Preprocessing, which integrates various morpheme forms. This transition showed an increase of up to 9.89% in the Weighted F1 score for the XGBoost classifier using the GloVe model. To verify the significance of this increase, we conducted paired t-tests between the most restrictive preprocessing, Root form + Detailed Preprocessing, and the more inclusive, Partial surface form + Partial Preprocessing. This latter form retains more morphological and syntactic information, potentially providing richer features for classifiers to utilize.

This statistical method is crucial for substantiating performance improvement claims by assessing whether observed differences in F1 scores are statistically significant or merely due to random variations. The paired t-test, conducted on the same sample sets, allows for a meticulous comparison of means by considering the variance within their differences. This provides a precise measure of whether performance discrepancies are coincidental or reflect a genuine effect. We set the significance threshold at $p < 0.05$, applying the tests to the average scores from two testing folds in a 10-fold cross-validation setup. This approach rigorously determines the significance of the performance differences between the compared models.

In evaluating sentiment classification performance across different embedding models, we strategically reduced the number of paired t-tests. This reduction is driven by the

TABLE 13. The Weighted F1 scores (%) for different non-contextual classification methods utilizing BoW embeddings processed through various preprocessing methods. The scores represent averages from two testing folds in a 10-fold cross-validation, highlighting how each preprocessing variant impacts the classification score.

Algorithms	Preprocessing Modules	Weighted F1
SVM	Root form + Detailed Preprocessing	47.62
	Root form + Partial Preprocessing	54.01
	Partial surface form + Partial Preprocessing	54.63
Decision Tree	Root form + Detailed Preprocessing	44.37
	Root form + Partial Preprocessing	48.42
	Partial surface form + Partial Preprocessing	49.61
Random Forest	Root form + Detailed Preprocessing	48.14
	Root form + Partial Preprocessing	49.92
	Partial surface form + Partial Preprocessing	50.02
Gradient Boosting	Root form + Detailed Preprocessing	41.49
	Root form + Partial Preprocessing	49.13
	Partial surface form + Partial Preprocessing	51.01
XGBoost	Root form + Detailed Preprocessing	47.1
	Root form + Partial Preprocessing	54.03
	Partial surface form + Partial Preprocessing	55.01
Multi Layer Perceptron	Root form + Detailed Preprocessing	47.09
	Root form + Partial Preprocessing	52.78
	Partial surface form + Partial Preprocessing	53.32

TABLE 14. The Weighted F1 scores (%) for various non-contextual classification methods utilizing TF-IDF feature embeddings. Scores are averaged from two testing folds within a 10-fold cross-validation, demonstrating the impact of different preprocessing approaches on classification performance.

Algorithms	Preprocessing Modules	Weighted F1
SVM	Root form + Detailed Preprocessing	46.84
	Root form + Partial Preprocessing	51.91
	Partial surface form + Partial Preprocessing	53.42
Decision Tree	Root form + Detailed Preprocessing	47.31
	Root form + Partial Preprocessing	50.35
	Partial surface form + Partial Preprocessing	50.39
Random Forest	Root form + Detailed Preprocessing	48.11
	Root form + Partial Preprocessing	49.71
	Partial surface form + Partial Preprocessing	51.16
Gradient Boosting	Root form + Detailed Preprocessing	43.41
	Root form + Partial Preprocessing	51.04
	Partial surface form + Partial Preprocessing	50.09
XGBoost	Root form + Detailed Preprocessing	45.12
	Root form + Partial Preprocessing	53.14
	Partial surface form + Partial Preprocessing	52.69
Multi Layer Perceptron	Root form + Detailed Preprocessing	47.85
	Root form + Partial Preprocessing	51.89
	Partial surface form + Partial Preprocessing	51.07

need to maintain statistical integrity and optimize resource use, focusing on the most substantial differences. Specific classifier and preprocessing combinations were selected based on their potential to demonstrate significant performance disparities. Multi-Layer Perceptron and XGBoost were chosen for their pronounced improvements when including morphological features and transitioning from more detailed to partial preprocessing settings. This selection is instrumental in showcasing the impact of preprocessing within the Non-Contextual Model framework, as detailed in Table 17.

The results from the paired t-tests evaluating the improvements in Weighted F1 scores when moving from Root form + Detailed Preprocessing to Partial surface form + Partial Preprocessing reveal a significant trend: less restrictive preprocessing, which preserves more morphological and syntactic features, generally results in substantial performance improvements in sentiment analysis models. This observation

holds across most classifier and embedding combinations, where improvements are marked by significant p-values ($p < 0.05$), indicating strong enhancements.

The results show that the most substantial improvement was observed with GloVe embeddings using XGBoost, where the “1st vs 3rd” comparison yielded a significant 9.89% increase in Weighted F1 scores ($p = 0.013$), demonstrating the effectiveness of this preprocessing strategy. However, some experiments, such as those involving TF-IDF embeddings with MLP ($p = 0.059$) and GloVe embeddings with MLP ($p = 0.053$), marginally met or exceeded the accepted p-value threshold of 0.05, indicating that the statistical significance was either borderline or not achieved. These findings suggest that the efficacy of preprocessing strategies may vary depending on the unique characteristics of the embedding models and their interaction with classifiers. Notably, the improvements with MLP using BoW embeddings were particularly significant ($p = 0.0024$), highlighting

TABLE 15. The weighted F1 scores (%) for various non-contextual classification methods utilizing Word2Vec embeddings. Scores are averaged from two testing folds within a 10-fold cross-validation, illustrating the effect of different preprocessing methods on classification efficacy.

Algorithms	Preprocessing Modules	Weighted F1
SVM	Root form + Detailed Preprocessing	46.32
	Root form + Partial Preprocessing	48.21
	Partial surface form + Partial Preprocessing	53.66
Decision Tree	Root form + Detailed Preprocessing	45.53
	Root form + Partial Preprocessing	51.04
	Partial surface form + Partial Preprocessing	49.91
Random Forest	Root form + Detailed Preprocessing	46.31
	Root form + Partial Preprocessing	50.45
	Partial surface form + Partial Preprocessing	52.92
Gradient Boosting	Root form + Detailed Preprocessing	45.92
	Root form + Partial Preprocessing	50.03
	Partial surface form + Partial Preprocessing	51.34
XGBoost	Root form + Detailed Preprocessing	45.53
	Root form + Partial Preprocessing	50.14
	Partial surface form + Partial Preprocessing	54.31
Multi Layer Perceptron	Root form + Detailed Preprocessing	43.87
	Root form + Partial Preprocessing	50.52
	Partial surface form + Partial Preprocessing	52.37

TABLE 16. The Weighted F1 scores (%) for various non-contextual classification methods utilizing GloVe embeddings. Scores are averaged from two testing folds within a 10-fold cross-validation, illustrating the impact of different preprocessing methods on classification efficacy.

Algorithms	Preprocessing Modules	Weighted F1
SVM	Root form + Detailed Preprocessing	47.11
	Root form + Partial Preprocessing	47.93
	Partial surface form + Partial Preprocessing	53.13
Decision Tree	Root form + Detailed Preprocessing	46.41
	Root form + Partial Preprocessing	49.43
	Partial surface form + Partial Preprocessing	49.42
Random Forest	Root form + Detailed Preprocessing	46.01
	Root form + Partial Preprocessing	50.76
	Partial surface form + Partial Preprocessing	53.23
Gradient Boosting	Root form + Detailed Preprocessing	47.31
	Root form + Partial Preprocessing	50.71
	Partial surface form + Partial Preprocessing	51.82
XGBoost	Root form + Detailed Preprocessing	44.83
	Root form + Partial Preprocessing	51.69
	Partial surface form + Partial Preprocessing	54.72
Multi Layer Perceptron	Root form + Detailed Preprocessing	45.16
	Root form + Partial Preprocessing	50.1
	Partial surface form + Partial Preprocessing	52.33

the advantage of incorporating a broader linguistic context in preprocessing.

f: PRE-TRAINED TURKISH BERT

In Table 18, we present the performance of supervised methods on pre-trained Turkish BERT utterance embeddings when ignoring the contextual, speaker, and inter-dependency of utterances. Both stopwords and punctuation marks may provide context about the sentiment of the subject; therefore, we did not eliminate them. We only performed basic preprocessing tasks for data preparation. As seen, we obtained the best sentiment classification score with the Multi-Layer Perceptron, which achieved 56.84%. When we compare the results in Table 18 with the results in Table 13, Table 14, Table 15, and Table 16, the Weighted F1 score of SVM, Decision Tree, Random Forest, and Gradient Boosting classifiers also outperformed the scores obtained using BoW,

TF-IDF, Word2Vec, and GloVe feature embeddings on the Non-Contextual Model.

In our Introduction section, we emphasized the exceptional performance of BERT-based utterance features over other vector embedding models within both non-contextual and contextual frameworks. To substantiate this for the non-contextual component, we conducted paired t-tests, specifically examining the weighted F1 scores obtained using the Multi-Layer Perceptron (MLP) across different embedding models. The MLP was selected due to its consistently superior performance when utilizing pre-trained Turkish BERT embeddings, making it an ideal candidate for this analysis.

The results of paired t-tests provided compelling statistical evidence of BERT's superiority. The tests, comparing the performance of BERT to that of Word2Vec, BoW, and TF-IDF under identical preprocessing configurations, yielded p-values of 0.0025, 0.0015, and 0.041 respectively. Each

TABLE 17. Paired T-Test results: The Weighted F1 scores of classifiers (MLP, XGBoost) across different preprocessing techniques. “1st vs 3rd” refers to the comparison between “Root form + Detailed Preprocessing” (1st) and “Partial surface form + Partial Preprocessing” (3rd), highlighting the impact of morpheme inclusion on sentiment analysis performance. The F1 Difference (%) shows the performance improvement, and the T-Test Value indicates the statistical significance, with p-values below 0.05 considered significant.

Embedding	Classifier	Comparison	Root form + Detailed Pre-processing	Partial surface form + Partial Preprocessing	F1 (%)	Difference	T-Test Value
TF-IDF	MLP	1st vs 3rd	47.85	51.07	+3.22		0.059
GloVe	MLP	1st vs 3rd	45.16	52.33	+7.17		0.053
TF-IDF	XGBoost	1st vs 3rd	45.12	52.69	+7.57		0.041
BoW	MLP	1st vs 3rd	47.09	53.32	+6.23		0.0024
BoW	XGBoost	1st vs 3rd	47.1	55.01	+7.91		0.023
Word2Vec	XGBoost	1st vs 3rd	45.53	54.31	+8.78		0.027
Word2Vec	MLP	1st vs 3rd	43.87	52.37	+8.50		0.0049
GloVe	XGBoost	1st vs 3rd	44.83	54.72	+9.89		0.013

TABLE 18. The Weighted F1 scores (%) for classifiers using BERT embeddings after basic preprocessing are shown. Scores from two testing folds in a 10-fold cross-validation demonstrate superior performance over traditional embeddings, with the multi-layer perceptron achieving the highest at 56.84%.

Classifier	Weighted F1
SVM	55.71
Decision Tree	54.12
Random Forest	55.49
Gradient Boosting	53.91
XGBoost	53.70
Multi-Layer Perceptron	56.84

of these values falls below the conventional significance threshold of 0.05, strongly supporting the superior capability of BERT in enhancing MLP’s ability to classify sentiment accurately and underscoring its effectiveness in capturing complex semantic relationships within the dataset.

However, the comparison between BERT and GloVe, conducted using the ‘Partial surface form + Partial Preprocessing’ setup for GloVe and the basic preprocessing form for BERT, resulted in a p-value of 0.057. This value, slightly above the standard significance threshold of 0.05, indicates a non-significant difference in performance under these differing conditions. BERT, processed with basic preprocessing, achieved an F1 score of 56.84%, while GloVe, under the more detailed ‘Partial surface form + Partial Preprocessing,’ scored 52.33%. It’s important to note that a p-value just above 0.05 does not conclusively indicate no performance difference between BERT and GloVe; rather, it suggests that the evidence is not robust enough to definitively reject the hypothesis of equal performance under the test conditions. This outcome implies that while the performance gap narrows with the specific preprocessing applied to GloVe, the difference does not reach statistical significance. This suggests that GloVe’s performance relative to BERT might be competitive under this detailed preprocessing but is not sufficient to claim equivalence or superiority.

Several factors underscore the superiority of BERT over traditional models. While TF-IDF and BoW lack contextual understanding, and GloVe and Word2Vec offer only a basic level of context comprehension, BERT stands out for its

ability to capture nuanced meanings across diverse contexts. BERT is built on the Transformer architecture, enabling it to learn intricate patterns and representations. It effectively handles out-of-vocabulary words and comprehends word morphemes, a capability absent in traditional models like TF-IDF and BoW. BERT embeddings not only capture semantic relationships between words but also encode syntactic information such as sentence structure and grammar, contributing to a deeper understanding of sentence meaning compared to Word2Vec. Given that BERT is pre-trained on an extensive dataset and has a larger number of parameters, it can capture effectively a broad spectrum of language patterns and knowledge.

2) DEEP LEARNING MODEL RESULTS

Table 19 displays the performance of a deep neural network architecture using pre-trained GloVe and pre-trained Turkish BERT model feature vectors. As explained and illustrated in the Non-Contextual Classifier section, the generated model is simplified. In contrast, DialogueRNN, generated using three GRUs to track individual speaker states, global context, and sentiment state during the conversation, introduces increased model complexity. Our intention was to observe and compare the success rates of our GloVe and pre-trained Turkish BERT embeddings when utilized as input in a simpler model, prior to being trained with a more complex one such as DialogueRNN.

The deep learning model, particularly the architecture designed with Keras that integrates BERT embeddings, demonstrates a distinct advantage as seen in Table 19. In the deep learning setup, BERT’s Weighted F1 score reaches 60.31%, outperforming GloVe’s 55.73%. This superior performance highlights BERT’s effectiveness in leveraging contextual clues and polysemy inherent in language, which is crucial for parsing the subtleties of sentiment in dialogue. The architecture, which includes a BERT layer that processes utterance embeddings with attention mechanisms to emphasize relevant contextual information, coupled with layers like GlobalMaxPooling1D and Dense layers, optimizes the neural network’s ability to discern and categorize sentiment more accurately than simpler embedding techniques such as GloVe.

TABLE 19. GloVe vs. pre-trained Turkish BERT embeddings performance: The Weighted F1 scores (%) for sentiment prediction using GloVe and BERT in the Non-Contextual Model. Scores are averaged from two testing folds in a 10-fold cross-validation, emphasizing BERT's superior ability to handle word polysemy and sequence context.

Feature	Weighted F1
GloVe	55.73
BERT	60.31

On the other hand, when BERT embeddings are utilized within traditional machine learning classifiers, as shown in Table 18, the results are robust but do not reach the performance levels observed with deep learning models. The highest score among these traditional classifiers is observed in the Multi-Layer Perceptron (MLP), achieving a Weighted F1 of 56.84%. Although MLP benefits from the rich representations provided by BERT, it lacks the complex sequential processing capabilities that deep learning models possess, which are crucial for effectively handling the dynamics of language used in couple dialogues. A paired t-test comparing the performance of deep learning models incorporating pre-trained Turkish BERT embeddings against MLP, which also utilized the pre-trained Turkish BERT model, yielded a significant result (p -value = 0.019), affirming the superiority of deep learning approaches under these testing conditions.

B. CONTEXTUAL CLASSIFIER RESULTS

1) DIALOGUERNN RESULTS

As previously outlined, we utilized a dual evaluation method for the DialogueRNN algorithm, comprising k -fold cross-validation and a standard training-validation-test split. Table 20 presents the 10-fold cross-validation results, with Weighted F1 scores averaged from two testing folds.

The DialogueRNN model incorporates various embeddings, including GloVe, pre-trained Turkish BERT, and text-embedding-3-small, each adapted for use in the DialogueRNN framework. These are respectively named GloVe-DIALOGUERNN, GPT-DIALOGUERNN, and Turkish BERT-DIALOGUERNN. Additionally, we have integrated embeddings from a version of the pre-trained Turkish BERT that has been specifically fine-tuned on Couple Dialogue data. This version is referred to as “Turkish BERT-DIALOGUERNN (Couple Dialogue Fine-tuned).”

a: EMBEDDING EFFECT

Table 20 shows that the integration of various embeddings in the DialogueRNN framework—specifically GloVe, GPT, and Turkish BERT—reveals distinct influences on sentiment analysis performance. GloVe-DIALOGUERNN demonstrated a lower Weighted F1 score of 52.36%, highlighting its limitations in capturing the dynamic nuances necessary for effective conversational sentiment analysis due to its context-independent nature. In contrast, GPT-DIALOGUERNN, which utilizes embeddings generated from a broad spectrum of

internet text via the OPENAI API, outperformed Turkish BERT-DIALOGUERNN with a score of 63.14%. This suggests that GPT's generative pre-training provides a more comprehensive language understanding, beneficial for processing the varied linguistic expressions found in couple dialogues.

Contextual embeddings like GPT and Turkish BERT significantly outperformed the less advanced GloVe model, underscoring the importance of context sensitivity in embeddings. These embeddings dynamically adjust their interpretations based on the surrounding text, offering a more nuanced understanding of dialogue intricacies. It is important to note that the paired t-test conducted between the Turkish BERT-DIALOGUERNN (Couple Dialogue Fine-tuned) and the GloVe-DIALOGUERNN revealed a statistically significant improvement ($p=0.05$), with a notable difference of 14.46% in performance. Notably, the fine-tuned Turkish BERT-DIALOGUERNN (Couple Dialogue Fine-tuned) showcased superior performance with the highest F1 score of 66.82%, illustrating the profound impact of domain-specific fine-tuning. By aligning the model more closely with the specific language patterns and sentiment expressions within the Couple Dialogue dataset, fine-tuning enhances the model's ability to interpret complex emotional cues effectively.

The DialogueRNN's ability to model inter-speaker dependencies and track conversational states is significantly enhanced by the advanced capabilities of these embeddings. This synthesis not only boosts the performance of sentiment classification tasks but also demonstrates the potential of combining sophisticated neural architectures with highly contextualized embeddings. The success of Turkish BERT-DIALOGUERNN, particularly after fine-tuning, underscores the benefit of embedding models that incorporate a deep understanding of both semantic and syntactic elements, proving essential in the evolving field of conversational AI.

b: CONTEXT EFFECT

The DialogueRNN model consistently outperforms Non-contextual Model experiments in conversational sentiment analysis, as evidenced by the comparative results in the provided tables (Table 20 versus Table 18 and Table 19). This advantage can primarily be attributed to DialogueRNN's ability to incorporate and dynamically update contextual information, speaker state, and previous sentiment. By maintaining stateful information across dialogues, DialogueRNN captures the flow and evolution of conversations more effectively than static, non-contextual models. This is particularly crucial in settings like couple dialogues, where the emotional undertone and sentiment can shift rapidly and are often influenced by the interaction history between speakers. The Contextual Model's superior performance underscores the importance of tailored architectures that align closely with the inherent structure of dialogic data, enabling a deeper understanding of sentiment dynamics within conversations.

We conducted paired t-tests to determine the significance of the performance improvements of the DialogueRNN

TABLE 20. The Weighted F1 scores (%) for sentiment prediction using GloVe, Turkish BERT, GPT, and fine-tuned Turkish BERT within DialogueRNN. Scores are averaged from two folds in a 10-fold cross-validation, highlighting differences in handling dialogue sentiment analysis.

Embedding Type	Weighted F1
Glove-DialogueRNN	52.36
Turkish BERT-DialogueRNN	62.16
GPT-DialogueRNN	63.14
Turkish BERT-DialogueRNN (Couple Dialogue Fine-tuned)	66.82

TABLE 21. The Weighted F1 scores (%) for different LLMs using methods like instruction-based prompting, few-shot prompting, and fine-tuning on Turkish couple dialogues. The results emphasize the superior accuracy of the Fine-Tuned Pre-Trained Turkish BERT, highlighting the benefits of language-specific tuning.

LLM Model	Weighted F1
Llama 2 with Instruction-based Prompting	34.49
Llama 2 with Few-shot Prompting	40.03
GPT-4 with Prompt Engineering	48.92
Llama 2 Fine-Tuning	54.67
GPT-3.5 Turbo with Fine-Tuning	65.42
Fine-Tuned Pre-Trained Turkish BERT	65.61

model over the Non-Contextual Model experiments, as shown in Tables 18 and Table 19. These tests were applied specifically to comparisons where the DialogueRNN model scores were higher. The first t-test compared the performance between the Turkish BERT-DialogueRNN (Couple Dialogue Fine-tuned) from the Contextual Model and the Multi-Layer Perceptron using pre-trained Turkish BERT from the Non-Contextual Model. The second t-test compared the Turkish BERT-DialogueRNN (Couple Dialogue Fine-tuned) from the Contextual Model to the deep learning model using pre-trained Turkish BERT from the Non-Contextual Model. The results of these tests yielded p-values of 0.0012 and 0.041, respectively, indicating statistically significant improvements.

2) LLM MODEL RESULTS

For the computationally demanding Large Language Models (LLMs), k-fold cross-validation was not feasible. Instead, we used a dataset split of 70% for training, 10% for validation, and 20% for testing to ensure consistent evaluation conditions. Table 21 presents the Weighted F1 scores from testing each LLM on the same test data, showcasing their performance across different setups.

a: FEW-SHOT PROMPTING VS. INSTRUCTION-BASED PROMPTING IN LLAMA 2

Table 21 demonstrates that Llama 2 with Few-shot Prompting outperforms its Instruction-based Prompting counterpart in conversational sentiment analysis on Couple Dialogue data, achieving a Weighted F1 score of 40.03% compared to 34.49%. This enhancement can be attributed to the few-shot model's ability to leverage multiple example interactions to

better understand the nuances of the task as illustrated in Table 7. Few-shot prompting effectively contextualizes the model's responses by providing it with clear examples of the desired output, helping it to adapt to the subtleties of positive, negative, and neutral emotional tones in conversations.

b: EFFECTIVENESS OF FINE-TUNING LLAMA 2

As demonstrated in Table 21, fine-tuning Llama 2 results in superior performance, achieving a Weighted F1 score of 54.67%, compared to 40.03% with few-shot prompting. Fine-tuning allows the model to adjust its neural network weights to better reflect the specific linguistic and sentiment patterns of the Couple Dialogue dataset. While prompt engineering directs the model's focus, fine-tuning reshapes its understanding, inherently making it more aligned with the dataset's unique characteristics. This adjustment promotes a deeper and more accurate recognition of sentiment cues, surpassing the surface-level comprehension achieved through prompt engineering.

c: PERFORMANCE OF LLAMA 2 ON TURKISH DATA

As discussed earlier, all Llama 2 tasks, including prompting and fine-tuning, have been conducted using the Trendyol LLM. This model has been fine-tuned with a substantial bilingual dataset that includes 180,000 instruction sets in both English and Turkish. Despite these efforts, the model's performance in sentiment analysis tasks for Turkish couple dialogues, as shown in Table 21, remained suboptimal. This might indicate that the proportion of Turkish examples in the training set does not adequately represent the linguistic diversity and complexity needed for our specific application. Although technically equipped to handle Turkish, the nuances and idiosyncrasies of conversational Turkish as manifested in couple dialogues may not be fully captured. The base Llama 2 model's predominance of English data training could still overshadow its capabilities in Turkish, leading to a performance gap when the language context is primarily Turkish.

d: LIMITATIONS OF GPT-4'S PROMPT ENGINEERING

GPT-4's performance in prompt engineering might seem underwhelming as shown in Table 21, particularly due to its exclusive reliance on instruction-based prompting, which lacks the richer contextual understanding provided by few-shot scenarios. Furthermore, if GPT-4's training included limited exposure to Turkish data, it would further impair its capability to process effectively and respond to the complexities of Turkish-language interactions. The absence of domain-specific fine-tuning also restricts the model's ability to adjust its learned behaviors to the specific nuances of the sentiment analysis task within the Turkish Couple Dialogue dataset.

Notably, despite GPT-3.5 Turbo being technologically less advanced than GPT-4, its superior performance post-fine-tuning highlights the importance of targeted training. We opted not to use few-shot prompting for GPT-4 primarily

due to budget constraints. The increased token requirements for adding few-shot examples would have tripled our costs with OPENAI, which was not feasible within our project budget. This limitation prevented us from leveraging the full potential of few-shot prompting, which might have enhanced GPT-4's performance in our specific application.

e: SUPERIORITY OF GPT-3.5 TURBO FINE-TUNING

The performance metrics presented in Table 21 highlight GPT-3.5 Turbo's superior efficacy in fine-tuning tasks compared to Llama 2, underscoring potential advantages in its architecture that are conducive to handling complex linguistic data, such as Turkish conversational sentiment analysis. Despite the lack of explicit architectural details due to proprietary constraints, the results from fine-tuning GPT-3.5 Turbo suggest that its underlying structure is responsive optimally to nuanced language features. This is likely due to the model's extensive pre-training across a diverse range of multilingual datasets, enabling it to acquire a substantial foundational understanding of language variations. When this broad base is fine-tuned to specific datasets, such as the Turkish Couple Dialogue, GPT-3.5 Turbo is able to adapt its learned patterns more effectively, enhancing its capacity to discern and process the intricate emotional dynamics within the dialogue. This adaptability and depth of learning, inferred from its performance, contribute to its notable success in this context, distinguishing it from Llama 2.

f: SUCCESS OF FINE-TUNED PRE-TRAINED TURKISH BERT

In the sentiment analysis of Turkish couple dialogues, as depicted in Table 21, the fine-tuned Pre-Trained Turkish BERT achieved the highest Weighted F1 score of 65.61%, surpassing both GPT-3.5 Turbo and Llama 2. This superior performance can be attributed to BERT's architecture, which is particularly adept at capturing the context within sentences and leveraging bidirectional training—distinct advantages when processing the complexities of conversational data. Unlike GPT models, which are trained unidirectionally and may not capture preceding context as effectively, BERT's bidirectional approach allows for a more nuanced understanding of the sentence structure and sentiment, adapting more precisely to the intricate language patterns found in Turkish dialogues. Additionally, the fine-tuning process on Turkish couple dialogue data further hones this model's ability to discern subtle emotional cues and linguistic intricacies unique to the dataset. This specialized adaptation to the linguistic context of the dialogue, combined with the inherent strengths of the BERT architecture, likely contributed to its outstanding performance, making it exceptionally well-suited for tasks requiring deep linguistic understanding and context sensitivity.

g: THE IMPACT OF CONTEXTUAL MODELING ON SENTIMENT ANALYSIS ACCURACY IN LLMs

The contrasting outcomes from the application of LLMs in contextual versus non-contextual frameworks for

conversational sentiment analysis on Couple Dialogue data clearly demonstrate the significance of context in enhancing model performance. As detailed in Table 19 and Table 21, the Fine-Tuned Pre-Trained Turkish BERT achieved the highest Weighted F1 score of 65.61% in the Contextual Model setup, surpassing all models in the non-contextual framework, where the highest score was 60.31% by BERT.

This significant enhancement is attributed to the Contextual Model's capability to utilize the immediate content and the broader conversational context, including historical interactions and speaker-specific nuances. These results underscore the superiority of contextual awareness in boosting the accuracy of sentiment analysis models. Although direct comparison is limited due to the use of different evaluation methods—fixed data splits for LLMs versus k-fold cross-validation for non-LLMs—and computational constraints that restricted the application of k-fold cross-validation for LLMs, the top-performing LLMs exhibit robust effectiveness. Specifically, even when utilizing only 20% of the data for testing during fine-tuning, models like the Fine-Tuned Pre-Trained Turkish BERT significantly outperformed other models. This highlights their potential under more extensive and rigorous testing frameworks. While some LLMs like the Llama 2 Fine-Tuning did not surpass Non-Contextual Model approaches, the overall higher scores among leading LLMs advocate for the benefits of contextual models in enhancing sentiment analysis. Further evaluations with statistical methodologies could substantiate these findings more comprehensively, yet the current results strongly advocate for the use of contextual models in enhancing sentiment analysis.

h: EVALUATING THE EFFICACY OF FINE-TUNING VS. PROMPT-BASED APPROACHES IN SENTIMENT ANALYSIS

Table 21 demonstrates that the fine-tuning approach significantly outperformed the prompt-based approach in our conversational sentiment analysis task, achieving an impressive improvement of up to 31.12% in the Weighted F1 score. Specifically, while the Llama 2 with Instruction-based Prompting yielded a Weighted F1 score of 34.49%, the score escalated to 65.61% with the Fine-Tuned Pre-Trained Turkish BERT. A paired t-test was conducted to determine the statistical significance of this change, resulting in a p-value of 0.037, under the standard threshold of 0.05. This confirms that the observed improvement is statistically significant, validating the superior performance of the fine-tuning approach over prompt-based methods in enhancing sentiment analysis accuracy.

i: SUPERIORITY OF DIALOGUERNN USING FINE-TUNED TURKISH BERT EMBEDDINGS

Table 22 presents the results of all Contextual Model approaches, including DialogueRNN, tested on a data split comprising 70% training, 10% validation, and 20% testing of the Couple Dialogue dataset. This setup facilitates

TABLE 22. The Weighted F1 scores for various Contextual Model algorithms tested with a standard dataset split of 70% training, 10% validation, and 20% testing. It highlights the comparative effectiveness of DialogueRNN and various large language models (LLMs) such as Llama 2, GPT-4, and Turkish BERT, with detailed performance metrics for different implementation strategies like prompting, fine-tuning, and DialogueRNN adaptations.

Contextual Model	Weighted F1
Llama 2 with Instruction-based Prompting	34.49
Llama 2 with Few-shot Prompting	40.03
GPT-4 with Prompt Engineering	48.92
Glove-DialogueRNN	52.01
Llama 2 Fine-Tuning	54.67
Turkish BERT-DialogueRNN	61.36
GPT-DialogueRNN	62.84
GPT 3.5 Turbo with Fine- Tuning	65.42
Fine-Tuned Pre-Trained Turkish BERT	65.61
Turkish BERT-DialogueRNN (Couple Dialogue Fine-tuned)	66.17

direct comparisons between DialogueRNN and various LLM models.

As illustrated in Table 22, the DialogueRNN model that integrates embeddings from the fine-tuned Turkish BERT, specifically customized for our Couple Dialogue dataset, achieved the highest performance with a Weighted F1 score of 66.17%. This configuration, labeled as Turkish BERT-DialogueRNN (Couple Dialogue Fine-tuned), leverages the fine-tuning process to closely align the model with the linguistic characteristics and sentiment nuances specific to our conversational data. The fine-tuning process adapts the pre-trained embeddings to better capture the contextual subtleties and emotional variances within the dialogue, offering more precise understanding and classification of sentiment. This is contrasted with the Turkish BERT-DialogueRNN, which uses general pre-trained embeddings and achieved a lower score of 61.36%, indicating the significant impact of domain-specific fine-tuning. Moreover, when compared to the Fine-Tuned Pre-Trained Turkish BERT (65.61%) and GPT-3.5 Turbo fine-tuning (65.42%), it is evident that embedding the fine-tuned model within the DialogueRNN framework enhances its capability to process and analyze the conversational dynamics more effectively, resulting in superior sentiment analysis performance.

The DialogueRNN framework is specifically designed to handle the complexities of sequential conversational data, effectively modeling the flow and dependencies of dialogue. By integrating fine-tuned embeddings, the DialogueRNN can leverage its architectural strengths to better utilize the contextual cues and speaker-specific information, which are crucial for accurate sentiment analysis in dialogue systems.

j: STATISTICAL SIGNIFICANCE AND CONFIDENCE INTERVAL ANALYSIS

Based on the statistical analysis and the results obtained in our study, we can confidently conclude that the contributions we outlined in the introduction are both valid and significant as shown in Table 23. The comparison between Non-Contextual

and Contextual Models clearly demonstrates the superiority of Contextual Model, with a statistically significant improvement of 9.98% in Weighted F1 scores when employing DialogueRNN with fine-tuned Turkish BERT embeddings. Similarly, the fine-tuned approach outperformed instruction-based prompting with a notable 31.12% increase in Weighted F1 score, further validated by a paired t-test. Our investigation into the integration of morphological features also revealed significant enhancements in classification performance, with improvements of up to 9.89% in Weighted F1 scores, particularly in the Non-Contextual Model where BERT embeddings showed an increase of 7.42%. Lastly, the BERT-based utterance features consistently outperformed other embedding models across both Non-Contextual and Contextual frameworks, with a marked 14.46% improvement in the Contextual Model. These findings are supported by robust statistical evidence, as reflected in the calculated t-values and confidence intervals, which provide a range within which the true difference in performance is likely to fall, offering additional assurance of the reliability of these findings.

VIII. DISCUSSION

At the start of this study, we defined several research questions to answer through our experiments. In this section, we propose an answer to each question and present possible options for future work. The research questions defined were the following:

- **RQ1:** Does the Contextual Model consistently outperform the Non-Contextual Model on the Couple Dialogue data?

The analytical synthesis of findings from contextual and non-contextual frameworks in our study highlights a definitive trend: Contextual Model approaches demonstrate a consistent superiority over their Non-Contextual counterparts in the realm of conversational sentiment analysis, particularly within the Couple Dialogue dataset. This trend is empirically underscored by the significant performance gap observable in the Weighted F1 scores, where Contextual Model leveraging dialogue-oriented embeddings and fine-tuning strategies—such as Turkish BERT-DialogueRNN that has been meticulously fine-tuned on couple dialogue data—outperform others. These models adeptly navigate the intricate emotional variations and flows characteristic of couple interactions by integrating a nuanced understanding of context, speaker dynamics, and sentiment progression. The evidence presented supports a clear conclusion: for sentiment analysis tasks rooted in dialogue data, Contextual Model approaches are not just advantageous but essential, enabling a more profound and coherent understanding of the subtleties inherent in conversational exchanges.

- **RQ2:** What is the most efficient prompt structure that would be understandable for both Llama 2 and GPT-3.5 Turbo in the context of conversational sentiment

TABLE 23. The statistical analysis of the Weighted F1 scores (%) across different settings, focusing on the significance of performance improvements. Comparisons are made between Non-Contextual and Contextual models, as well as different preprocessing techniques. The F1 difference (%) reflects the improvement in performance, while the T-Test value assesses statistical significance, with p-values below 0.05 considered significant. Confidence intervals (CI) are calculated based on a 95% confidence level. Abbreviations used: "DNN" refers to a deep neural network architecture incorporating Pre-trained Turkish BERT, "IBP" refers to instruction-based prompting.

Comparison Type	Model 1	Model 1 Weighted F1	Model 2	Model 2 Weighted F1	F1 Difference (%)	T-Test Value	CI
Non-Contextual vs. Non-Contextual	Decision Tree (Partial Surface + Partial Preprocessing, GloVe)	49.42	MLP (Pre-trained Turkish BERT)	56.84	+7.42	0.0023	[3.43, 11.41]
Non-Contextual vs. Non-Contextual	XGBoost (Root Form + Detailed Preprocessing, GloVe)	44.83	XGBoost (Partial Surface + Partial Preprocessing, GloVe)	54.72	+9.89	0.013	[2.64, 17.14]
Non-Contextual vs. Contextual	MLP (Pre-trained Turkish BERT)	56.84	Turkish BERT-DialogueRNN (Couple Dialogue Fine-tuned)	66.82	+9.98	0.0012	[5.13, 14.83]
Non-Contextual vs. Contextual	DNN (Pre-trained Turkish BERT)	60.31	Turkish BERT-DialogueRNN (Couple Dialogue Fine-tuned)	66.82	+6.51	0.041	[0.33, 12.69]
Contextual vs. Contextual	Llama 2 (IBP)	34.49	Fine-Tuned Turkish BERT	65.61	+31.12	0.037	[2.34, 59.90]
Contextual vs. Contextual	GloVe-DialogueRNN	52.36	Turkish BERT-DialogueRNN (Couple Dialogue Fine-tuned)	66.82	+14.46	0.0032	[6.24, 22.68]

analysis on the Couple Dialogue data? (This question arises because, to the best of our knowledge, a suitable prompt structure for this task has not been documented in the literature.)

The exploration of prompt efficacy for Llama 2 and GPT-3.5 Turbo in the context of conversational sentiment analysis on Couple Dialogue data reveals that Few-shot Prompting provides a more effective framework than Instruction-based Prompting. This is evidenced by the superior performance of Llama 2 with Few-shot Prompting, which harnesses the power of example-driven learning to understand and categorize emotional tones within dialogues. The efficient prompt structure, therefore, would be one that integrates multiple examples of dialogues with clearly annotated emotional sentiments, as seen in the Few-shot Prompting approach.

Such a prompt structure benefits LLMs by providing concrete instances of the task at hand, allowing for a better grasp of the nuanced differences between positive, negative, and neutral sentiments as they manifest in natural conversation. It also guides the models to discern and apply conversational dynamics and speaker gender considerations more adeptly. Consequently, to ensure that both Llama 2 and GPT-3.5 Turbo can effectively conduct sentiment analysis in Turkish couple dialogues, the prompts should include several annotated examples that illustrate the expected analysis. This approach not only facilitates a deeper understanding of the emotional layers in the conversations but also primes the models to generate more accurate sentiment classifications in line with the provided examples.

- **RQ3:** Can our conversational sentiment analysis task, which is based on the Turkish language, be effectively achieved by Llama 2 and GPT-3.5 Turbo, both of which are primarily pre-trained in English?

The capability of Llama 2, even when adapted as the Trendyol LLM fine-tuned on bilingual data, to handle effectively the subtleties of Turkish conversational sentiment analysis remains limited, suggesting that the nuances of Turkish dialogue may not be entirely within its grasp. On the other hand, GPT-3.5 Turbo exhibits a more nuanced understanding and adaptability to the complexities of Turkish, which can be attributed to its expansive pre-training over a diverse linguistic corpus. The fine-tuning process, particularly with a dataset as specialized as the Turkish Couple Dialogue, enables GPT-3.5 Turbo to finely tune its recognition of the language's unique emotional contours, offering more effective sentiment analysis in this context. This suggests that while both models hold promise, the success of sentiment analysis tasks in Turkish is more feasible with GPT-3.5 Turbo, given its advanced adaptability and learning depth.

- **RQ4:** How much improvement do fine-tuned models offer over prompt-based approaches for this conversational sentiment analysis task?
Fine-tuned models offer substantial improvement over prompt-based approaches in conversational sentiment analysis, particularly in linguistically complex scenarios such as analyzing Turkish couple dialogues. This improvement is primarily due to the fine-tuning process's ability to adapt models to the specific linguistic and contextual nuances of the target dataset. In contrast to prompt-based methods, which may struggle with capturing the full depth of conversational dynamics due to their reliance on less customized inputs, fine-tuning leverages the entire conversational context. This allows models to better understand and respond to the intricate patterns of dialogue, including shifts in tone and sentiment that are critical in such analyses.

- **RQ5:** Do contextual embeddings outperform non-contextual vector representations like BoW and TF-IDF, as well as word embeddings with minimal contextual understanding such as GloVe and Word2Vec, for tasks involving Couple Dialogue data?

Contextual embeddings, such as those derived from BERT or GPT, significantly outperform other vector representations like BoW, TF-IDF, Word2Vec, and GloVe in conversational sentiment analysis tasks for the Couple Dialogue data. BERT's architecture, built on the Transformer model, excels in understanding and utilizing the context of sentences, capturing the nuances of language that are often missed by non-contextual methods. These embeddings are particularly adept at dealing with complex language patterns and sentence structures, which are crucial in understanding the sentiments expressed in couple dialogues. Moreover, BERT embeddings integrate syntactic and semantic information, enabling a deeper and more accurate analysis of sentiment compared to traditional vector representations.

- **RQ6:** What are the effects of preprocessing and morphological analysis on a language with rich morphology, like Turkish, in enhancing sentiment labeling success rates? Preprocessing and morphological analysis play a crucial role in enhancing sentiment labeling success rates for languages with rich morphology like Turkish. The inclusion of morphological details and preprocessing adjustments significantly impacts the effectiveness of sentiment analysis classifiers. By incorporating stopwords and specific punctuation, classifiers are able to better understand the informal nuances and emotional expressions typically found in conversational Turkish, particularly in contexts like couple dialogues. The use of morpheme-specific flags, which highlight emotional, sentimental, or negational aspects within words, enhances the ability of models to discern subtle cues within the language, thereby improving sentiment classification accuracy. Such preprocessing not only preserves essential linguistic elements but also aligns more closely with the natural use of language, allowing for more accurate and nuanced sentiment analysis. This approach demonstrates the importance of tailored preprocessing strategies in handling the complexities of languages enriched with morphological variations, ultimately leading to better model performance and deeper linguistic insights.
- **RQ7:** Does the use of a more complex neural network architecture in the Non-Contextual Model help close the performance gap with the Contextual Model? The adoption of complex neural network architectures in the Non-Contextual Model significantly narrows the performance gap with the Contextual Model, especially in conversational sentiment analysis. Complex

architectures, like those using BERT embeddings and attention mechanisms, enhance the model's ability to process language nuances effectively, extracting and utilizing contextual clues and polysemy crucial for accurate sentiment interpretation. These advanced models, with layers such as GlobalMaxPooling1D and Dense layers, are better equipped to discern subtle emotional variations in dialogue than simpler models, leading to notable improvements in sentiment classification accuracy. However, despite these advancements, they still do not fully reach the effectiveness of fully the Contextual Model approaches like DialogueRNN, which is specially designed to manage the dynamics of conversations by tracking individual speaker states and evolving context, thus offering a deeper understanding and higher accuracy in conversational analysis.

IX. CONCLUSION AND FUTURE WORK

This study's exploration into the "Couple Dialogue" dataset, a pioneering initiative in the realm of Turkish conversational sentiment analysis, has underscored the indispensable role of contextual modeling in decoding the complexities of human interaction. The comparative analysis between Non-Contextual and Contextual Models reveals that contextual approaches, particularly those employing advanced Large Language Models like BERT, GPT-3.5 Turbo, Llama 2, and GPT-4, significantly enhance the accuracy of sentiment analysis. By incorporating state-of-the-art techniques and architectures such as DialogueRNN, which adeptly processes the relational and sequential dynamics of dialogues, our research has achieved a notable 9.98% improvement in Weighted F1 scores over traditional Non-contextual Model approaches. This success not only demonstrates the effectiveness of the Contextual Model approaches in capturing the subtle nuances and emotional transitions within conversations but also establishes a robust benchmark for future sentiment analysis endeavors within linguistically rich contexts.

The rigorous testing and analysis across various modeling frameworks have further confirmed that contextual embeddings profoundly outperform non-contextual vector representations like BoW, TF-IDF, and basic word embeddings such as GloVe and Word2Vec. These findings validate the superiority of contextual methods in handling the intricate language patterns and emotional subtleties embedded in the Turkish Couple Dialogue data. Our study highlights the critical importance of fine-tuning and the use of sophisticated prompt structures in enhancing the performance of LLMs, making it possible to capture the nuanced sentiment expressions inherent in couple dialogues more effectively. As such, the Contextual Model has not only outperformed its Non-Contextual counterpart consistently but has also set a new standard for conducting sentiment analysis in complex conversational datasets.

A. CHALLENGES IN CONVERSATIONAL SENTIMENT ANALYSIS IN COUPLE DIALOGUE DATA

Analyzing emotional dynamics through conversational sentiment analysis presents considerable challenges, necessitating sophisticated approaches to modeling conversational contexts. Our dataset, in Turkish and specifically pertaining to couple interactions, cannot be directly compared with other datasets using the DialogueRNN architecture. Translating our dataset into English does not preserve the original meaning due to lengthy and complex utterances. As shown in [112], and emphasized by recent research, translation accuracy, cultural nuances, and idiomatic expressions present significant hurdles. These challenges include the loss of subtle linguistic cues and cultural contexts, which machine translation systems often miss, leading to inaccuracies in sentiment analysis outcomes. Idioms and colloquial expressions often do not translate directly, requiring advanced techniques for accurate sentiment interpretation across linguistic barriers. Additionally, the absence of comparable studies on conversational sentiment analysis in Turkish leaves us without a benchmark. A significant challenge we faced was the failure to detect sarcasm effectively within our Couple Dialogue dataset, which is informal and centers on couple interactions.

In Table 24, we present the utterance predictions from a script derived from the 20% testing set, following a standard dataset split of 70% for training, 10% for validation, and 20% for testing. The script includes dialogues with a sarcastic theme, which are labeled as “0” (negative) by human annotators. Our analysis of the Couple Dialogue dataset reveals that the Turkish BERT-DialogueRNN (Couple Dialogue Fine-tuned) model exhibited the highest performance in terms of Weighted F1 score, as detailed in Table 22. Despite its overall effectiveness in sentiment analysis, this model, along with other contextual model approaches, faced challenges in accurately classifying sarcastic expressions. These challenges are evident in Table 24, where the nuanced and context-dependent nature of sarcasm becomes apparent. Our experimental results indicate that GPT-based models, despite their general effectiveness, struggled with accurately classifying sarcastic expressions in the Turkish language. This is evident from the performance metrics and examples provided in Table 24. The nuanced and context-dependent nature of sarcasm presents a significant challenge, and the models showed limited success in this regard. This observation highlights a gap in the current capabilities of these models to fully interpret the subtle cues necessary for sarcasm detection, which often depends heavily on cultural and contextual insights beyond the literal text.

B. COMPREHENSIVE STRATEGIES FOR ENHANCING SARCASM DETECTION

The challenge of detecting sarcasm using text-based models underscores the necessity of incorporating additional modalities, such as audio data, to capture the full spectrum

of communicative cues more effectively. Vocal tone and intonation are pivotal in conveying sarcasm and can greatly enhance model performance when integrated. We have begun to collect audio data from couple interactions, planning to merge this with our existing models to better assess and improve their ability to detect sarcasm.

Additionally, our dataset includes detailed annotations for both sentiment and emotional tones, which provides a robust foundation for training our models. Integrating these annotations will allow our models to better understand the emotional context of utterances, enhancing their capability to discern not just the literal meaning but also the emotional subtext, which is essential for accurately identifying sarcasm.

Looking ahead, further advancements in sarcasm detection within the complex linguistic landscape of Turkish couple dialogues can be achieved through sophisticated Large Language Model (LLM) techniques. Specifically, models like BERT and GPT, which have been pre-trained on extensive multilingual corpora, will be fine-tuned using task-specific datasets that encompass examples of sarcasm. Employing strategies such as transfer learning to adapt these models for Turkish, and utilizing few-shot learning to accommodate the nuanced detection of sarcasm with minimal training examples, will likely bridge significant gaps in current model performance. By integrating these advanced LLM techniques with multimodal data inputs, we anticipate these models will become significantly more adept at detecting sarcasm, thereby enhancing the accuracy and reliability of sentiment analysis outcomes in conversational settings.

C. ADVANCED LLM APPLICATIONS IN REAL-TIME SENTIMENT ANALYSIS AND PRACTICAL USE CASES

Our research highlights the crucial role of advanced Large Language Models (LLMs) such as GPT-3.5 Turbo, Llama 2, BERT, and GPT-4 in significantly improving the accuracy and depth of sentiment analysis across various sectors. These models enhance the capabilities of digital assistants in customer service by enabling them to interpret and respond to complex customer emotions swiftly and empathetically. For instance, deploying these LLMs within customer service bots in the telecommunications sector has resulted in reduced response times and increased customer satisfaction through more contextually aware interactions.

Real-time sentiment analysis applications, particularly in customer support systems, present further opportunities to improve user experiences. Integrating our contextual sentiment analysis models into live platforms can enhance the immediacy and relevance of virtual agents' responses, which is especially beneficial for customer service centers handling high volumes of interactions. Immediate sentiment analysis can direct queries to appropriate channels, thereby improving operational efficiency and customer satisfaction. Research by Zhu [65] on emotion recognition-based smart assistants underscores the importance of emotional intelligence in enhancing user interaction, aligning with our findings that

TABLE 24. This table from the couple dialogue dataset provides examples of sarcasm within a dialogue script, illustrating the predictions by all the Contextual Model experiments used in our conversational sentiment analysis. “M” denotes male and “F” denotes female speakers. Sentimental shifts are annotated with “2” for positive, “0” for negative, and “1” for neutral. The table columns display predictions from each Contextual Model approach, while the “True Sentiment” column shows the human-labeled true sentiment. Abbreviations are used for model names to conserve space, including IBP (Instruction-based prompting), FSP (Few-shot prompting), PE (Prompt engineering), FT (Fine-tuning), DRNN (DialogueRNN), and PT (Pre-trained).

Speaker	Utterance (Turkish and English Translation)	True Sentiment	Llama 2 IBP	Llama 2 FSP	GPT- 4 PE	GloVe- DRNN	Llama 2 FT	Turkish BERT- DRNN	GPT- DRNN	GPT 3.5 Turbo FT	Fine- Tuned Pre- Trained Turkish BERT	Turkish BERT- DRNN (Couple Dia- logue Fine- tuned)
F	Oh, çamaşırlar kendi kendine makineye girecek mi sanıyorsun? Ben atmasam bir yere gitmezler! (Oh, do you think the laundry will just walk itself to the washer? If I don't put them in, they're not going anywhere!)	0	1	1	1	1	1	1	1	0	0	0
F	Neden sadece kadınların işi gibi ev işleri, hep oturup kraliçe gibi yaşamıyor muyum zaten? (Why are house chores like only a woman's job? Am I not already living like a queen sitting around?)	0	2	2	1	2	1	0	1	1	0	0
M	Evlilikte mutluluk mu? Ha, evet, herkes için ama öncelikle senin için tabii. (Marriage and happiness? Ha, yes, for everyone, but first and foremost for you, of course.)	0	2	1	1	1	1	1	0	1	1	0
F	Tabii, çamaşır dolabını her gün selamlayıp konuşuyorum, sen de görebilirsin aslında! (Of course, I greet and talk to the laundry cabinet every day, you could see it too if you tried!)	0	2	2	2	2	2	1	2	0	1	1
M	Çoraplar bitti, belki ayaklarım donar, belki o zaman anlarsın. (Socks are finished, maybe my feet will freeze, maybe then you'll understand.)	0	1	0	0	0	0	0	0	0	0	0
M	Çamaşır sepetine at, tabii canım, sanki başka işim yokmuş gibi. (Put it in the laundry basket, sure dear, as if I have nothing else to do.)	0	1	1	1	1	1	1	1	0	1	1
F	Çamaşır makinesine atamazsan büyük bir trajedi olacak, hayat duracak. (If you don't put them in the washing machine, it'll be a great tragedy, life will stop.)	0	1	0	0	1	0	0	0	0	0	0
M	Çıkarıp oraya atıyorum, evet, tam bir hizmetçi gibiyim. (I'm taking it out and putting it there, yes, just like a servant.)	0	1	1	1	1	1	0	1	1	0	0

real-time sentiment analysis can lead to more responsive and personalized services.

However, deploying these models in real-time settings presents challenges, including the need for robust computational infrastructure and effective data streaming techniques to manage high-volume, high-velocity data. The system must process and analyze data instantaneously to ensure timely and accurate responses, requiring setups that maintain both stability and performance while handling dynamic data streams.

Building on this discussion, our Couple Dialogue dataset serves as a valuable resource for testing and validating sentiment analysis models in conditions that closely approximate real-world scenarios. Although the data were collected in a controlled laboratory environment, the interactions themselves are highly naturalistic, reflecting the genuine, unscripted nature of marital discussions. The dataset was meticulously gathered from a diverse sample of 103 recently married heterosexual couples in Turkey, each engaging in conversations about personally significant, contentious topics. This setting allowed for the capture of dynamic interactions, complete with the “noise” typical of everyday communication—such as overlapping speech, informal language, and unstructured dialogue patterns.

While the dataset is not a perfect replica of real-world conditions, as it was collected in a lab setting, the realism in the conversations, combined with the careful transcription and annotation, ensures that it closely mirrors the unpredictability and conversational nuances found in everyday life. This positions the Couple Dialogue dataset as a strong, though not without its limitations, resource for evaluating the robustness of sentiment analysis models in practical applications. It highlights the potential challenges these models might face when deployed in truly noisy, real-world environments, where the unpredictability and complexity of human communication are at their peak.

D. CROSS-DOMAIN INTEGRATION FOR ADVANCED HEALTHCARE SENTIMENT ANALYSIS

Our next initiative aims to leverage the advanced sentiment analysis capabilities of the latest Large Language Models (LLMs), including Llama 3.1 and GPT-4o. GPT-4o, a faster and more capable variant of GPT-4, excels in benchmarks such as simple math, language comprehension, and vision understanding, according to OpenAI.⁹ Meanwhile, Meta Llama 3.1, as recently unveiled by Meta,¹⁰ offers improvements like increased context length, expanded tool use, and enhanced multilingual capabilities. Our project will deploy these models within the healthcare domain, specifically analyzing psychiatrist-patient therapy sessions for conditions like depression and Alzheimer’s disease.

Building on the sentiment detection techniques developed for the “Couple Dialogue” dataset, this project will track

emotional shifts during therapy to help psychiatrists monitor and assess therapeutic progress. Ethical permissions for data collection and analysis have been secured for this project, which will be conducted in Turkish at a public hospital in Turkey. Upon completion of data collection, we plan to apply our refined Contextual Model approaches, with detailed results to be shared in an upcoming publication.

We also see potential in integrating this psychiatric session data with our existing Couple Dialogue dataset. Since both datasets are in Turkish, our current models can be applied seamlessly, enriching the training process and improving the detection of complex emotional dynamics across different conversational contexts. This approach will test the adaptability of our models in mental health evaluations, aiming to refine their precision in understanding patient states.

This cross-domain application not only validates the versatility of our sentiment analysis models but also extends their use into critical areas of mental health. By supporting more nuanced and effective therapeutic interventions, we aim to enhance the robustness and accuracy of our models, potentially making them essential tools in clinical psychology and psychiatry, thereby influencing therapeutic strategies and patient care.

E. EXPLORING CROSS-LINGUISTIC MODELS: A MULTIFACETED APPROACH

1) CHALLENGES AND OPPORTUNITIES

Building on our established foundation in sentiment analysis with contextual embeddings tailored to the Turkish language, we seek to expand our research through the exploration of cross-linguistic models. These models utilize extensive multilingual datasets, allowing sentiment analysis without the need for language-specific models. This capability enhances utility in global customer service and social media platforms but also introduces challenges such as varying sentiment expressions across cultures, word ambiguities, and syntactic differences, which can significantly impact model accuracy and context comprehension.

2) EXPLORING MULTILINGUAL MODEL PERFORMANCE AND FUTURE DIRECTIONS FOR ENHANCED CROSS-LINGUISTIC UNDERSTANDING

While advanced models like LLaMA 2, GPT-4, and GPT-3.5 Turbo exhibit cross-linguistic capabilities, they predominantly emphasize English, which can limit their effectiveness in capturing the nuances of other languages like Turkish. As demonstrated in Table 22, Turkish BERT, pre-trained exclusively on Turkish data, is more specialized and effective for tasks in the Turkish language. Although fine-tuning generalist models like GPT-3.5 Turbo on the Couple Dialogue dataset improved their performance (e.g., GPT-3.5 Turbo with Fine-Tuning scoring 65.42), these models still fell short compared to the specialized Turkish BERT. This highlights the potential advantage of pre-training on a specific language dataset to achieve superior performance in that language.

⁹<https://openai.com/index/hello-gpt-4o/>

¹⁰<https://huggingface.co/meta-llama/Meta-Llama-3.1-8B>

To address the limitations observed with models predominantly trained on English data, we plan to fine-tune the Couple Dialogue dataset using mBERT [113] and XLM-R [114]. These models are optimized for multilingual processing and are better suited to handle the structural complexities of Turkish. This future work will assess whether these models can surpass the limitations of their generalist counterparts and improve cross-linguistic understanding, particularly in the context of Turkish sentiment analysis.

3) SYNTHESIZING MULTILINGUAL DATA SOURCES

We plan to enhance our sentiment analysis capabilities by merging our Turkish Couple Dialogue dataset with an English dataset of couple dialogues. This integration will optimize the LLaMA 3.1 model's performance across both languages through fine-tuning, better capturing each language's unique linguistic nuances.

Finding open-source datasets specifically focused on dyadic couple dialogues poses a challenge, so we have requested couples therapy session data from various authors. Additionally, IEMOCAP [7], a valuable open-source resource, features emotional dyadic interactions through acted dialogues, closely resembling real couple interactions. To supplement real-world data, we will leverage the latest advancements in generative models like GPT-4o, building on our previous success with GPT-4 in creating synthetic data.

Combining datasets from Turkish and English offers significant advantages. Turkish's intricate morphological structure contrasts with English's simpler syntax, providing a valuable comparative framework. The abundance of resources and benchmarks in English enhances model training, facilitating more effective cross-linguistic transfer. Utilizing advanced pre-trained models like GPT-4o and LLaMA 3.1 allows for rigorous benchmarking and further refinement, expanding the training landscape and improving the models' ability to process and generate language across diverse linguistic contexts, thereby significantly increasing their versatility and robustness.

APPENDIX A

DISTINGUISHING BETWEEN SIMILAR EMOTIONS

A. EXAMPLE 1: FEAR VS. SURPRISE

- **Dialogue Snippet:** “Ne yaptın? Gece oraya yalnız mı gittin?” / “You did what? You went there alone at night?”
- **Context:** Reaction to someone recounting an unexpected solo visit to a known dangerous area.
- **Annotation Challenge:** Deciding whether the emotion is surprise at the action taken or fear for the person's safety.
- **Guideline Application:** The follow-up concern about safety indicates fear.
- **Final Sentiment:** Given the safety concern, the sentiment is **Negative**.

B. EXAMPLE 2: SADNESS VS. NEUTRAL

- **Dialogue Snippet:** “Eskisi gibi değiliz.” / “We are not like we used to be.”
- **Context:** Said during a calm discussion reflecting on relationship changes.
- **Annotation Challenge:** Determining whether the statement expresses sadness about changes or a neutral observation.
- **Guideline Application:** If the tone conveys a sense of loss, it indicates sadness.
- **Final Sentiment:** If reflective of loss, the sentiment is **Negative**; if merely observational, it is **Neutral**.

C. EXAMPLE 3: HAPPINESS VS. SURPRISE

- **Dialogue Snippet:** “Sana sürprizim var!” / “I have a surprise for you!”
- **Context:** Said during a special moment like a birthday.
- **Annotation Challenge:** Deciding whether this is an expression of genuine happiness or just surprise.
- **Guideline Application:** The context and tone indicating joy suggest genuine happiness.
- **Final Sentiment:** If joy and happiness are evident, the sentiment is **Positive**; if it merely points to an unexpected situation, it could be **Neutral**.

APPENDIX B

EXAMPLES OF THE EMOTIONAL BLENDS

A. EXAMPLE 1: HAPPINESS VS. SADNESS

- **Dialogue Snippet:** “Bugün terfi aldım, çok mutluyum!” / “I got promoted today, I'm so happy!” “Ama bu terfiyle birlikte daha az zaman geçireceğiz.” / “But with this promotion, we'll spend less time together.”
- **Context:** Said during a conversation about a recent promotion.
- **Annotation Challenge:** Distinguishing between genuine happiness about the promotion and sadness about the potential impact on their time together.
- **Guideline Application:** The first sentence clearly shows happiness due to the promotion (label as Happiness), while the second sentence reflects sadness about the decrease in time spent together (label as Sadness).
- **Final Sentiment:** If the sadness about time spent together is more deeply felt and emphasized in subsequent context, the final label should be **Sadness** (Negative). Otherwise, if the happiness outweighs the concern, it can be **Happiness** (Positive).

B. EXAMPLE 2: POSITIVE SURPRISE VS. NEGATIVE FEAR

- **Dialogue Snippet:** “Bana yeni bir araba aldın mı? Bu inanılmaz!” / “Did you buy me a new car? This is amazing!” “Ama bu kadar para harcadığından korkuyorum.” / “But I'm afraid of how much money you spent.”
- **Context:** Said during a surprise reveal of a new car.

Algorithm 1 Complete Pseudocode of the Conversational Sentiment Analysis Study on the Couple Dialogue Dataset. This Listing Encompasses All Computational Methods Employed in the Analysis

```

1: Input: CoupleDialogueDataset
2: Output: SentimentLabels, WeightedF1Score
3:
4: NonContextualFeatureMethods  $\leftarrow$  [BoW, TF-IDF, Word2Vec, GloVe, PretrainedTurkishBERT]
5: ContextualFeatureMethods  $\leftarrow$  [GloVe, PretrainedTurkishBERT, FineTunedTurkishBERT, OpenAITextEmbedding]
6:
7: Define Non-Contextual and Contextual classifiers:
8: NonContextualClassifiers  $\leftarrow$  [DecisionTree, RandomForest, GradientBoosting, XGBoost, SVM,
9:   MultiLayerPerceptron, DeepLearningModel(Keras)]
10: ContextualClassifiers  $\leftarrow$  [PromptBased(GPT-4, Llama2), FineTuned(BERT, Llama2, GPT3.5Turbo), DialogueRNN]
11:
12: Define Basic Preprocessing:
13: BasicPreprocessing  $\leftarrow$  { "ISO encoding:" True, "expanding contractions:" True, "eliminating numerical values:" True,
14:   "removing single characters:" True, "removing excessive spaces:" True, "lowercasing:" True }
15:
16: Define Additional Preprocessing Configurations:
17: PreprocessingConfigs  $\leftarrow$  {
18:   "DetailedRootForm:" BasicPreprocessing + { "remove_stopwords:" True, "remove_punctuation:" True,
19:     "retain_specific_punctuations:" [] },
20:   "PartialRootForm:" BasicPreprocessing + { "remove_stopwords:" False, "remove_punctuation:" True,
21:     "retain_specific_punctuations:" ["!", "..."] },
22:   "SurfaceForm:" BasicPreprocessing + { "remove_stopwords:" False, "remove_punctuation:" True,
23:     "retain_specific_punctuations:" ["!", "..."], "include_morphemes:" True },
24:   "BasicPreprocessing:" BasicPreprocessing
25: }
26:
27: function initialize_nested_dictionary
28:   return lambda: defaultdict(initialize_nested_dictionary)
29:
30: function TrainAndEvaluateNonContextualModels(Dataset, NonContextualFeatureMethods, PreprocessingConfigs)
31:   NonContextualScoresByPreprocessing  $\leftarrow$  initialize_nested_dictionary()
32:   for each ConfigName, ConfigDetails in PreprocessingConfigs do
33:     for each FeatureMethod in NonContextualFeatureMethods do
34:       for each Classifier in NonContextualClassifiers do
35:         if Classifier == DeepLearningModel(Keras) and (FeatureMethod == "GloVe"
36:           or FeatureMethod == "PretrainedTurkishBERT") then
37:
38:           PreprocessedDataset  $\leftarrow$  ApplyPreprocessing(Dataset, PreprocessingConfigs["BasicPreprocessing"])
39:           ConfigName  $\leftarrow$  "BasicPreprocessing"
40:
41:         else if FeatureMethod == "PretrainedTurkishBERT" and Classifier != DeepLearningModel(Keras) then
42:           PreprocessedDataset  $\leftarrow$  ApplyPreprocessing(Dataset, PreprocessingConfigs["BasicPreprocessing"])
43:           ConfigName  $\leftarrow$  "BasicPreprocessing"
44:
45:         else
46:           PreprocessedDataset  $\leftarrow$  ApplyPreprocessing(Dataset, ConfigDetails)
47:           FoldScores  $\leftarrow$  initialize_empty_list()
48:
49:           for each Fold in KFold(PreprocessedDataset, 10) do
50:             Train, Validation, Test  $\leftarrow$  Fold
51:             TrainFeatures, ValidationFeatures, TestFeatures  $\leftarrow$  ExtractFeatures(Train, Validation, Test, FeatureMethod)
52:             OptimizedHyperparameters  $\leftarrow$  OptimizeHyperparameters(ValidationFeatures, Classifier)
53:             TestScore  $\leftarrow$  EvaluateModel(TestFeatures, Classifier, OptimizedHyperparameters)
54:             FoldScores.Append(TestScore)
55:             AverageFoldScore  $\leftarrow$  Average(FoldScores)
56:             NonContextualScoresByPreprocessing[ConfigName][FeatureMethod][Classifier]  $\leftarrow$  AverageFoldScore
57:   return NonContextualScoresByPreprocessing
58:
59: function FineTuneModel(Model, TrainData, ValidationData)
60:   Model  $\leftarrow$  InitializeModel(Model)
61:   for each Epoch in TrainingEpochs do
62:     Model.Train(TrainData)
63:     ValidationScore  $\leftarrow$  Model.Evaluate(ValidationData)
64:   return Model, ValidationScore

```

Algorithm 2 Complete Pseudocode of the Conversational Sentiment Analysis Study on the Couple Dialogue Dataset. This Listing Encompasses All Computational Methods Employed in the Analysis. (Continued)

```

1: function EvaluateDialogueRNNWithKFold(Dataset, ContextualFeatureMethods)
2:   KFoldResultsByEmbedding  $\leftarrow$  initialize_empty_dictionary()
3:   for each EmbeddingType in ContextualFeatureMethods do
4:     KFoldResults  $\leftarrow$  initialize_empty_list()
5:     for each Fold in KFold(Dataset, 10) do
6:       Train, Validation, Test  $\leftarrow$  Fold
7:       FoldTrainFeatures  $\leftarrow$  ExtractDialogueFeatures(Train, EmbeddingType)
8:       FoldValidationFeatures  $\leftarrow$  ExtractDialogueFeatures(Validation, EmbeddingType)
9:       FoldTestFeatures  $\leftarrow$  ExtractDialogueFeatures(Test, EmbeddingType)
10:      OptimizedHyperparameters  $\leftarrow$  OptimizeHyperparameters(FoldValidationFeatures)
11:      TestScore  $\leftarrow$  EvaluateModel(FoldTestFeatures, DialogueRNN, OptimizedHyperparameters)
12:      KFoldResults.Append(TestScore)
13:      FoldAverageScore  $\leftarrow$  Average(KFoldResults)
14:      KFoldResultsByEmbedding[EmbeddingType]  $\leftarrow$  FoldAverageScore
15:   return KFoldResultsByEmbedding
16:
17: function EvaluateDialogueRNNStandardSplit(TrainData, ValidationData, TestData, NonContextualFeatureMethods)
18:   StandardScoresByEmbedding  $\leftarrow$  initialize_empty_dictionary()
19:   for each EmbeddingType in NonContextualFeatureMethods do
20:     DialogueFeaturesTrain  $\leftarrow$  ExtractDialogueFeatures(TrainData, EmbeddingType)
21:     DialogueFeaturesValidation  $\leftarrow$  ExtractDialogueFeatures(ValidationData, EmbeddingType)
22:     DialogueFeaturesTest  $\leftarrow$  ExtractDialogueFeatures(TestData, EmbeddingType)
23:     OptimizedHyperparameters  $\leftarrow$  OptimizeHyperparameters(DialogueFeaturesValidation)
24:     DialogueRNNStandardScore  $\leftarrow$  EvaluateModel(DialogueFeaturesTest, DialogueRNN, OptimizedHyperparameters)
25:     StandardScoresByEmbedding[EmbeddingType]  $\leftarrow$  DialogueRNNStandardScore
26:   return StandardScoresByEmbedding
27:
28: function TrainAndEvaluateContextualModels(TrainData, ValidationData, TestData)
29:   ContextualScores  $\leftarrow$  initialize_empty_dictionary()
30:   for Classifier in ContextualClassifiers.PromptBased do
31:     if Classifier is GPT-4 then
32:       ApplyInstructionBasedPrompting(Classifier, TrainData, ValidationData)
33:     else if Classifier is Llama2 then
34:       ApplyInstructionBasedPrompting(Classifier, TrainData, ValidationData)
35:       ApplyFewShotPrompting(Classifier, TrainData, ValidationData)
36:       Score  $\leftarrow$  EvaluateModel(TestData, Classifier)
37:       ContextualScores[Classifier]  $\leftarrow$  Score
38:   for FineTunedClassifier in ContextualClassifiers.FineTuned do
39:     Model, Score  $\leftarrow$  FineTune(FineTunedClassifier, TrainData, ValidationData)
40:     Score  $\leftarrow$  EvaluateModel(TestData, Model)
41:     ContextualScores[FineTunedClassifier]  $\leftarrow$  Score
42:   return ContextualScores
43:
44: DatasetSplit  $\leftarrow$  StandardSplit(CoupleDialogueDataset)
45:
46: NonContextualResults  $\leftarrow$  TrainAndEvaluateNonContextualModels(CoupleDialogueDataset, NonContextualFeatureMethods,
47:   PreprocessingConfigs)
48:
49: DialogueRNNKFoldResults  $\leftarrow$  EvaluateDialogueRNNWithKFold(CoupleDialogueDataset, ContextualFeatureMethods)
50: DialogueRNNStandardResults  $\leftarrow$  EvaluateDialogueRNNStandardSplit(DatasetSplit.TrainData, DatasetSplit.ValidationData,
51:   DatasetSplit.TestData, ContextualFeatureMethods)
52:
53: ContextualResults  $\leftarrow$  TrainAndEvaluateContextualModels(DatasetSplit.TrainData, DatasetSplit.ValidationData, DatasetSplit.TestData)
54:
55: NonContextualVsDialogueRNNComparison  $\leftarrow$  CompareF1Scores(NonContextualResults, DialogueRNNKFoldResults)
56: ContextualModelComparisons  $\leftarrow$  CompareAllContextualModels(ContextualResults, DialogueRNNStandardResults)
57:
58: Print("Non-Contextual Classifier Scores by Preprocessing:," NonContextualResults)
59: Print("DialogueRNN K-Fold F1 Scores by Embedding:," DialogueRNNKFoldResults)
60: Print("Contextual Classifier Scores:," ContextualResults)
61: Print("Non-Contextual vs DialogueRNN K-Fold Comparison:," NonContextualVsDialogueRNNComparison)
62: Print("Contextual Model Comparisons (Standard Split):," ContextualModelComparisons)

```

TABLE 25. Software packages used in the study.

Package Name	Description	Version
openai	Provides access to AI models like GPT.	0.28.0
torch	ML library for vision and NLP.	2.3.0
tensorflow	Framework for numerical computation.	2.16.1
keras	Neural networks API, runs on TensorFlow.	3.3.3
pandas	Data manipulation and analysis library.	2.2.1
numPy	Supports large, multi-dimensional arrays.	1.26.4
scikit-learn	Supports supervised and unsupervised learning.	1.4.1
transformers	Pretrained models for text tasks.	4.40.1
nltk	Toolkit for human language data.	3.8.1
spaCy	Industrial-strength NLP in Python.	3.7.4
tqdm	Progress bar for Python and CLI.	4.66.2
statsmodels	Explore data, estimate models, and perform tests.	0.14.0
datasets	Accessing and sharing datasets in AI.	2.18.0
seaborn	Statistical graphics.	0.13.2
matplotlib	Plotting library for Python and NumPy.	3.8.3
peft	Allows for the fine-tuning of large language models without the need to update all of the model's parameters.	0.9.0
trl	Hugging Face's library for supervised fine-tuning (SFT); which is very useful for training LLMs on instruction datasets.	0.7.12
tensorboard	Visualization tool for TensorFlow, providing metrics plot, graphs display, and more.	2.16.2
gensim	Gensim is a Python library for topic modeling and semantic analysis using NLP.	4.3.1
Microsoft Office Spell Checker	Microsoft Office's Spell Checker for Turkish language provides spelling, grammar, and style corrections, along with contextual suggestions and the ability to add custom words to enhance writing accuracy and clarity.	2016

- **Annotation Challenge:** Distinguishing between the initial positive surprise and the subsequent negative fear regarding the financial implications.
- **Guideline Application:** The first sentence demonstrates positive surprise with excitement and amazement (label as Surprise), while the second sentence introduces negative fear about the financial cost (label as Fear).

- **Final Sentiment:** If the fear about the financial cost is more pressing and highlighted in the following utterances, the final label should be **Fear** (Negative). If the surprise and joy dominate the conversation, it can be **Surprise** (Positive).

C. EXAMPLE 3: DISGUST VS. NEUTRAL

- **Dialogue Snippet:** “Bu yemek çok kötü kokuyor, bunu yiyemem!” / “This food smells awful, I can’t eat this!” “Ama belki sadece benim damak tadına uymuyordur.” / “But maybe it just doesn’t suit my taste.”
- **Context:** Said during a meal where the food is found unappealing.
- **Annotation Challenge:** Differentiating between strong disgust and a neutral assessment.
- **Guideline Application:** The first sentence shows strong disgust with vivid language (label as Disgust), while the second sentence provides a neutral explanation (label as Neutral).
- **Final Sentiment:** If the disgust is more intense and repeated in the conversation, the final label should be **Disgust** (Negative). If the neutral assessment prevails and reduces the intensity of disgust, it can be **Neutral**.

APPENDIX C PSEUDOCODE

The complete pseudocode of the conversational sentiment analysis on the Couple Dialogue dataset is given in Algorithm

APPENDIX D SOFTWARE PACKAGES

Table 25 presents information on the packages used in this study, including a short description and the versions installed.

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